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Imaging lens system, image capturing unit and electronic device

Abstract

An imaging lens system includes eight lens elements which are, in order from an object side to an image side: a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element. Each of the eight lens elements has an object-side surface facing toward the object side and an image-side surface facing toward the image side. At least one lens element of the imaging lens system has at least one lens surface having at least one inflection point. The imaging lens system has a total of eight lens elements.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation patent application of U.S. application Ser. No. 17/580,507, filed on Jan. 20, 2022, which is a continuation patent application of U.S. application Ser. No. 16/371,220, filed on Apr. 1, 2019, which claims priority to Taiwan Application 107137600, filed on Oct. 24, 2018, which is incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates to an imaging lens system, an image capturing unit and an electronic device, more particularly to an imaging lens system and an image capturing unit applicable to an electronic device.

Description of Related Art

(2) With the development of semiconductor manufacturing technology, the performance of image sensors has been improved, and the pixel size thereof has been scaled down. Therefore, featuring high image quality becomes one of the indispensable features of an optical system nowadays.

(3) Furthermore, due to the rapid changes in technology, electronic devices equipped with optical systems are developed towards multi-functionality for various applications, and therefore the functionality requirements for the optical systems have been increasing. However, it is yet difficult

for a conventional optical system to obtain a balance among the requirements such as high image quality, low sensitivity, proper aperture size, miniaturization or a desirable field of view.

Accordingly, there is a need for an optical system that satisfies the aforementioned requirements.

SUMMARY

(4) According to one aspect of the present disclosure, an imaging lens system includes eight lens elements. The eight lens elements are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element. Each of the eight lens elements has an object-side surface facing toward the object side and an image-side surface facing toward the image side. The first lens element has positive refractive power. At least one lens element of the imaging lens system has at least one lens surface having at least one inflection point. The imaging lens system has a total of eight lens elements. When a minimum value among Abbe numbers of all lens elements of the imaging lens system is V_{min} , half of a maximum field of view of the imaging lens system is HFOV, a focal length of the imaging lens system is f , a focal length of the first lens element is f_1 , an entrance pupil diameter of the imaging lens system is EPD, an axial distance between the image-side surface of the eighth lens element and an image surface is BL, and an axial distance between the object-side surface of the first lens element and the image-side surface of the eighth lens element is TD, the following conditions are satisfied:

$$8.0 < V_{min} < 22.5;$$

$$5.0[\text{deg.}] < \text{HFOV} < 25.0[\text{deg.}];$$

$$0.80 < f/EPD < 2.20;$$

$$0 < BL/TD < 0.25; \text{ and}$$

$$0.32 < f/f_1 < 3.80.$$

(5) According to another aspect of the present disclosure, an image capturing unit includes the aforementioned imaging lens system and an image sensor, wherein the image sensor is disposed on the image surface of the imaging lens system.

(6) According to still another aspect of the present disclosure, an electronic device includes at least two image capturing units, and the at least two image capturing units face the same side. The at least two image capturing units include a first image capturing unit and a second image capturing unit. The first image capturing unit includes the aforementioned imaging lens system and a first image sensor, and the first image sensor is disposed on the image surface of the imaging lens system. The second image capturing unit includes an optical lens assembly and a second image sensor, and the second image sensor is disposed on an image surface of the optical lens assembly. A field of view of the first image capturing unit and a field of view of the second image capturing unit are different from each other by at least 20 degrees.

(7) According to yet another aspect of the present disclosure, an imaging lens system includes eight lens elements. The eight lens elements are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element. Each of the eight lens elements has an object-side surface facing toward the object side and an image-side surface facing toward the image side. At least one lens element of the imaging lens system has at least one lens surface having at least one inflection point. The imaging lens system has a total of eight lens elements. When an Abbe number of the eighth lens element is V_8 , half of a maximum field of view of the imaging lens system is HFOV, a focal length of the imaging lens system is f , an entrance pupil diameter of the imaging lens system is EPD, and an axial distance between the object-side surface of the first lens element and an image surface is TL, the following conditions are satisfied:

$$8.0 < V_8 < 24.5;$$

$$5.0[\text{deg.}] < \text{HFOV} < 30.0[\text{deg.}];$$

$$0.80 < f/EPD < 2.20; \text{ and}$$

$$0.50 < TL/f < 1.30.$$

(8) According to yet still another aspect of the present disclosure, an imaging lens system includes eight lens elements. The eight lens elements are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element. Each of the eight lens elements has an object-side surface facing toward the object side and an image-side surface facing toward the image side. At least one lens element of the imaging lens system has at least one lens surface having at least one inflection point. The imaging lens system has a total of eight lens elements. When a minimum value among Abbe numbers of all lens elements of the imaging lens system is V_{min} , an axial distance between the object-side surface of the first lens element and an image surface is TL , a focal length of the imaging lens system is f , an axial distance between the image-side surface of the eighth lens element and the image surface is BL , and an axial distance between the object-side surface of the first lens element and the image-side surface of the eighth lens element is TD , the following conditions are satisfied:

$$8.0 < V_{min} < 20.0;$$

$$0.50 < TL/f < 1.10; \text{ and}$$

$$0 < BL/TD < 0.30.$$

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be better understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

(2) FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure;

(3) FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment;

(4) FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure;

(5) FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment;

(6) FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure;

(7) FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment;

(8) FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure;

(9) FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment;

(10) FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure;

(11) FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment;

(12) FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure;

(13) FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment;

(14) FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure;

(15) FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the

image capturing unit according to the 7th embodiment;

(16) FIG. 15 is a schematic view of an image capturing unit according to the 8th embodiment of the present disclosure;

(17) FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment;

(18) FIG. 17 is a schematic view of an image capturing unit according to the 9th embodiment of the present disclosure;

(19) FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment;

(20) FIG. 19 is a schematic view of an image capturing unit according to the 10th embodiment of the present disclosure;

(21) FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 10th embodiment;

(22) FIG. 21 is a schematic view of an image capturing unit according to the 11th embodiment of the present disclosure;

(23) FIG. 22 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 11th embodiment;

(24) FIG. 23 is a schematic view of an image capturing unit according to the 12th embodiment of the present disclosure;

(25) FIG. 24 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 12th embodiment;

(26) FIG. 25 is a schematic view of an image capturing unit according to the 13th embodiment of the present disclosure;

(27) FIG. 26 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 13th embodiment;

(28) FIG. 27 is a perspective view of an image capturing unit according to the 14th embodiment of the present disclosure;

(29) FIG. 28 is one perspective view of an electronic device according to the 15th embodiment of the present disclosure;

(30) FIG. 29 is another perspective view of the electronic device in FIG. 28;

(31) FIG. 30 is a block diagram of the electronic device in FIG. 28; and

(32) FIG. 31 shows a schematic view of Y11, Y51, Yc72, Y82 and an inflection point and a critical point on the image-side surface of the seventh lens element according to the 1st embodiment of the present disclosure.

DETAILED DESCRIPTION

(33) An imaging lens system includes eight lens elements. The eight lens elements are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element.

(34) The first lens element can have positive refractive power; therefore, it is favorable for providing significant light converging capability so as to minimize the space between adjacent lens elements, thereby achieving compactness. The first lens element can have an object-side surface being convex in a paraxial region thereof; therefore, it is favorable for receiving light rays from the off-axis region so as to reduce the incident angle of light, thereby preventing total reflection.

(35) The second lens element can have negative refractive power; therefore, it is favorable for correcting aberrations generated by the first lens element so as to eliminate spherical and chromatic aberrations. The second lens element can have an image-side surface being concave in a paraxial region thereof; therefore, it is favorable for further correcting aberrations so as to improve image quality.

(36) The third lens element can have an image-side surface being concave in a paraxial region

thereof. Therefore, it is favorable for providing a compact and telephoto configuration while reducing aberrations.

(37) The seventh lens element can have an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for reducing the back focal length of the imaging lens system so as to meet the requirement of compactness.

(38) According to the present disclosure, at least one lens element of the imaging lens system has at least one lens surface having at least one inflection point; therefore, it is favorable for correcting off-axis aberrations and reducing the size of the imaging lens system. Moreover, each of at least three lens elements of the imaging lens system can have at least one lens surface having at least one inflection point; therefore, it is favorable for correcting coma and astigmatism, obtaining a compact configuration and flattening the Petzval surface of the imaging lens system. Please refer to FIG. 31, which shows a schematic view of an inflection point P on the image-side surface 172 of the seventh lens element 170 according to the 1st embodiment of the present disclosure. The inflection point P on the image-side surface of the seventh lens element in FIG. 31 is only exemplary. The other lens surfaces of the eight lens elements may also have one or more inflection points.

(39) When a minimum value among Abbe numbers of all lens elements of the imaging lens system is $V_{\min}$, the following condition can be satisfied: $8.0 < V_{\min} < 22.5$. Therefore, it is favorable for adjusting the light path and balancing the capability of converging light rays with different wavelengths so as to correct chromatic aberration. Moreover, the following condition can be satisfied: $8.0 < V_{\min} < 20.0$. Moreover, the following condition can be satisfied: $10.0 < V_{\min} < 20.0$. Moreover, the following condition can also be satisfied: $15.0 < V_{\min} < 19.0$.

(40) When half of a maximum field of view of the imaging lens system is HFOV, the following condition can be satisfied: $5.0 [\text{deg.}] < \text{HFOV} < 30.0 [\text{deg.}]$. Therefore, it is favorable for controlling the field of view so as to obtain a better imaging range, and make the imaging lens system applicable to various applications. Moreover, the following condition can also be satisfied: $5.0 [\text{deg.}] < \text{HFOV} < 25.0 [\text{deg.}]$.

(41) When a focal length of the imaging lens system is f , and an entrance pupil diameter of the imaging lens system is EPD, the following condition can be satisfied: $0.80 < f/\text{EPD} < 2.20$. Therefore, it is favorable for adjusting the entrance pupil so as to provide sufficient incident light to increase image brightness. Moreover, the following condition can be satisfied: $1.0 < f/\text{EPD} < 2.0$. Moreover, the following condition can also be satisfied: $0.80 < f/\text{EPD} < 1.80$.

(42) When an axial distance between an image-side surface of the eighth lens element and an image surface is BL, and an axial distance between the object-side surface of the first lens element and the image-side surface of the eighth lens element is TD, the following condition can be satisfied: $0 < \text{BL}/\text{TD} < 0.30$. Therefore, it is favorable for reducing the back focal length so as to control the total track length of the imaging lens system. Moreover, the following condition can be satisfied: $0 < \text{BL}/\text{TD} < 0.25$. Moreover, the following condition can also be satisfied: $0 < \text{BL}/\text{TD} < 0.18$.

(43) When the focal length of the imaging lens system is f , and a focal length of the first lens element is f_1 , the following condition can be satisfied: $0.32 < f/f_1 < 3.80$. Therefore, it is favorable for providing sufficient converging capability on the object side of the imaging lens system from the first lens element so as to achieve a telephoto configuration.

(44) When an Abbe number of the eighth lens element is V_8 , the following condition can be satisfied: $8.0 < V_8 < 24.5$. Therefore, it is favorable for correcting chromatic aberration by the eighth lens element so as to prevent defocusing of light rays with different wavelengths, thereby improving image quality. Moreover, the following condition can be satisfied: $8.0 < V_8 < 22.0$. Moreover, the following condition can also be satisfied: $8.0 < V_8 < 20.0$.

(45) When an axial distance between the object-side surface of the first lens element and the image surface is TL, and the focal length of the imaging lens system is f , the following condition can be satisfied: $0.50 < \text{TL}/f < 1.30$. Therefore, it is favorable for balancing between the total length and the field of view of the imaging lens system, while making the imaging lens system applicable to

various applications. Moreover, the following condition can also be satisfied: $0.50 < TL/f < 1.10$.

(46) According to the present disclosure, each of at least four lens elements of the imaging lens system can be made of plastic material. Therefore, it is favorable for reducing manufacturing costs and increasing design flexibility of the imaging lens system so as to optimize the capability for correcting off-axis aberrations.

(47) When the axial distance between the object-side surface of the first lens element and the image surface is TL , and the entrance pupil diameter of the imaging lens system is EPD , the following condition can be satisfied: $0.90 < TL/EPD < 1.90$. Therefore, it is favorable for achieving a short total track length and a large aperture, enabling the imaging lens system to capture high brightness images within limited interior space. Moreover, the following condition can also be satisfied: $1.10 < TL/EPD < 1.67$.

(48) According to the present disclosure, each of at least four lens elements of the imaging lens system can have an Abbe number smaller than 35.0. Therefore, it is favorable for providing the materials of the lens elements in the imaging lens system with a sufficient capability of controlling light path so as to balance the focusing positions of light rays with different wavelengths, thereby preventing image overlapping. Moreover, each of at least four lens elements of the imaging lens system can have an Abbe number smaller than 30.0. Moreover, each of at least four lens elements of the imaging lens system can have an Abbe number smaller than 25.0. According to the present disclosure, the Abbe number V of one lens element is obtained from the following equation: $V = (N_d - 1)/(N_F - N_C)$, wherein N_d is the refractive index of said lens element at the wavelength of helium d-line (587.6 nm), N_F is the refractive index of said lens element at the wavelength of hydrogen F-line (486.1 nm), and N_C is the refractive index of said lens element at the wavelength of hydrogen C-line (656.3 nm).

(49) When a focal length of the fifth lens element is f_5 , and a focal length of the eighth lens element is f_8 , the following condition can be satisfied: $-0.50 < f_8/f_5 < 9.0$. Therefore, it is favorable for providing sufficient capability of controlling light path with proper refractive power of the eighth lens element and controlling the refractive power among different fields of view in the middle part of the imaging lens system so as to reduce aberrations. Moreover, the following condition can also be satisfied: $-0.40 < f_8/f_5 < 3.0$.

(50) When a sum of central thicknesses of all lens elements of the imaging lens system is ΣCT , and a sum of axial distances between each of all adjacent lens elements of the imaging lens system is ΣAT , the following condition can be satisfied: $1.10 < \Sigma CT/\Sigma AT < 3.50$. Therefore, it is favorable for balancing the thicknesses of the lens elements and gaps between adjacent lens elements of the imaging lens system so as to achieve a high assembling yield rate. Moreover, the following condition can also be satisfied: $1.20 < \Sigma CT/\Sigma AT < 2.50$.

(51) When a curvature radius of the object-side surface of the first lens element is R_1 , and a central thickness of the first lens element is CT_1 , the following condition can be satisfied: $0.50 < R_1/CT_1 < 2.65$. Therefore, it is favorable for enhancing the capability of controlling light path and the structural strength of the first lens element so as to allow the imaging lens system to be applicable to different environmental conditions. Moreover, the following condition can also be satisfied: $0.80 < R_1/CT_1 < 2.50$.

(52) When a curvature radius of an object-side surface of the seventh lens element is R_{13} , and a curvature radius of the image-side surface of the seventh lens element is R_{14} , the following condition can be satisfied: $(R_{13} - R_{14})/(R_{13} + R_{14}) < 5.0$. Therefore, it is favorable for balancing the surface shapes of the seventh lens element so as to better correct aberrations, thereby improving image quality. Moreover, the following condition can be satisfied: $-2.0 < (R_{13} - R_{14})/(R_{13} + R_{14}) < 3.0$. Moreover, the following condition can also be satisfied: $-1.0 < (R_{13} - R_{14})/(R_{13} + R_{14}) < 2.0$.

(53) When a maximum effective radius of the object-side surface of the first lens element is Y_{11} , a maximum effective radius of the image-side surface of the eighth lens element is Y_{82} , a maximum image height of the imaging lens system (half of a diagonal length of an effective photosensitive

area of an image sensor) is ImgH , and the focal length of the imaging lens system is f , the following condition can be satisfied: $0 < (|Y11 - Y82| + |Y82 - \text{ImgH}|) / f < 0.15$. Therefore, it is favorable for balancing the differences among the object side, the image side and the image height of the imaging lens system, and controlling the diameters of the lens elements so as to maintain the quality in lens molding, thereby providing a stable manufacturing yield rate. Please refer to FIG. 31, which shows a schematic view of $Y11$ and $Y82$ according to the 1st embodiment of the present disclosure.

(54) When a maximum value among refractive indices of all lens elements of the imaging lens system is N_{max} , the following condition can be satisfied: $1.60 < N_{\text{max}} < 1.72$. Therefore, it is favorable for selecting proper lens materials so as to reduce difficulty in manufacture, thereby increasing the feasibility of commercialization of the imaging lens system. Moreover, the following condition can also be satisfied: $1.65 < N_{\text{max}} < 1.70$.

(55) When the focal length of the imaging lens system is f , a curvature radius of an object-side surface of one lens element of the imaging lens system is R_o , and a curvature radius of an image-side surface of the lens element is R_i , at least one lens element of the imaging lens system can satisfy the following condition: $|f/R_o| + |f/R_i| < 0.50$. Therefore, it is favorable for providing at least one correction lens in the imaging lens system and preventing the curvature of lens surfaces thereof from being overly large so that the imaging lens system is able to balance aberrations generated by the lens elements on the object side and image side. Moreover, at least one lens element of the imaging lens system can satisfy the following condition: $|f/R_o| + |f/R_i| < 0.30$.

(56) When the entrance pupil diameter of the imaging lens system is EPD , and the axial distance between the image-side surface of the eighth lens element and the image surface is BL , the following condition can be satisfied: $4.50 < \text{EPD}/\text{BL} < 18.0$. Therefore, it is favorable for maintaining a proper back focal length for lens elements assembling within limited interior space, and providing sufficient amount of incident light so as to meet the requirement of product specifications. Moreover, the following condition can also be satisfied: $4.50 < \text{EPD}/\text{BL} < 9.0$.

(57) When the axial distance between the image-side surface of the eighth lens element and the image surface is BL , and the central thickness of the first lens element is CT1 , the following condition can be satisfied: $0 < \text{BL}/\text{CT1} < 0.95$. Therefore, it is favorable for increasing the thickness of the first lens element for having sufficient structural strength of the imaging lens system against environmental factors so as to maintain consistent product quality; furthermore, it is favorable for controlling the back focal length of the imaging lens system so as to achieve compactness. Moreover, the following condition can also be satisfied: $0.10 < \text{BL}/\text{CT1} < 0.70$.

(58) When the maximum effective radius of the object-side surface of the first lens element is $Y11$, the maximum image height of the imaging lens system is ImgH , and the entrance pupil diameter of the imaging lens system is EPD , the following condition can be satisfied: $-0.27 < (Y11 - \text{ImgH})/\text{EPD} < 0$. Therefore, it is favorable for balancing the differences between the object side and the image side of the imaging lens system and providing sufficient amount of incident light.

(59) When the focal length of the imaging lens system is f , and the focal length of the fifth lens element is f_5 , the following condition can be satisfied: $-0.25 < f/f_5 < 1.50$. Therefore, it is favorable for balancing the refractive power of the fifth lens element so as to reduce sensitivity of the imaging lens system. Moreover, the following condition can also be satisfied: $0 < f/f_5 < 1.0$.

(60) According to the present disclosure, the imaging lens system further includes an aperture stop. When an axial distance between the aperture stop and the image surface is SL , and the axial distance between the object-side surface of the first lens element and the image surface is TL , the following condition can be satisfied: $0.50 < \text{SL}/\text{TL} < 0.95$. Therefore, the position of the aperture stop is favorable for controlling the size of the imaging lens system. Moreover, the following condition can also be satisfied: $0.50 < \text{SL}/\text{TL} < 0.85$.

(61) When the focal length of the imaging lens system is f , half of the maximum field of view of

the imaging lens system is HFOV, and the entrance pupil diameter of the imaging lens system is EPD, the following condition can be satisfied: $0.20 < f \times \tan(\text{HFOV}) / \text{EPD} < 1.0$. Therefore, it is favorable for obtaining a telephoto configuration with sufficient incident light so as to prevent low image brightness. Moreover, the following condition can also be satisfied: $0.30 < f \times \tan(\text{HFOV}) / \text{EPD} < 0.73$.

(62) When the maximum effective radius of the object-side surface of the first lens element is **Y11**, the maximum effective radius of the image-side surface of the eighth lens element is **Y82**, the axial distance between the image-side surface of the eighth lens element and the image surface is BL, and the focal length of the imaging lens system is f, the following condition can be satisfied: $0 < (|Y11 - Y82| + BL) / f < 0.20$. Therefore, it is favorable for preventing an overly large difference between pupil openings at two ends of the imaging lens system to provide sufficient incoming light, and for reducing the back focal length so as to obtain a proper space arrangement of the imaging lens system.

(63) When an Abbe number of the first lens element is **V1**, an Abbe number of the second lens element is **V2**, an Abbe number of the third lens element is **V3**, an Abbe number of the fourth lens element is **V4**, an Abbe number of the fifth lens element is **V5**, an Abbe number of the sixth lens element is **V6**, an Abbe number of the seventh lens element is **V7**, the Abbe number of the eighth lens element is **V8**, and an Abbe number of the i-th lens element is **Vi**, the following condition can be satisfied: $150.0 < \sum Vi < 320.0$, wherein $i=1, 2, 3, 4, 5, 6, 7, 8$. Therefore, it is favorable for increasing the density difference between the lens elements and air so as to better control the light path within limited interior space. Moreover, the following condition can also be satisfied: $150.0 < \sum Vi < 300.0$, wherein $i=1, 2, 3, 4, 5, 6, 7, 8$.

(64) When the focal length of the imaging lens system is f, the focal length of the fifth lens element is **f5**, and a focal length of the sixth lens element is **f6**, the following condition can be satisfied: $-0.50 < f / f5 + f / f6 < 1.0$. Therefore, it is favorable for correcting aberrations with proper refractive power of the fifth lens element and sixth lens element so as to optimize the image quality. Moreover, the following condition can also be satisfied: $-0.20 < f / f5 + f / f6 < 0.50$.

(65) When a vertical distance between a critical point on the image-side surface of the seventh lens element and an optical axis is **Yc72**, and the focal length of the imaging lens system is f, the following condition can be satisfied: $0.02 < Yc72 / f < 0.70$. Therefore, it is favorable for correcting off-axis aberrations on the image side while reducing distortion and field curvature. Moreover, the following condition can also be satisfied: $0.02 < Yc72 / f < 0.28$. Please refer to FIG. 31, which shows a schematic view of a critical point C on the image-side surface **172** of the seventh lens element **170** according to the 1st embodiment of the present disclosure.

(66) When the focal length of the fifth lens element is **f5**, and a focal length of the seventh lens element is **f7**, the following condition can be satisfied: $f5 / f7 < 0.50$. Therefore, it is favorable for balancing the refractive power of the fifth lens element and seventh lens element so as to better correct aberrations, thereby improving image quality. Moreover, the following condition can also be satisfied: $-5.0 < f5 / f7 < 0.35$.

(67) When a curvature radius of an image-side surface of the fourth lens element is **R8**, and the focal length of the imaging lens system is f, the following condition can be satisfied: $-1.50 < R8 / f < 1.50$. Therefore, it is favorable for providing sufficient refraction on the image-side surface of the fourth lens element so as to control the light path. Moreover, the following condition can also be satisfied: $0 < R8 / f < 1.50$.

(68) When the focal length of the imaging lens system is f, and the maximum image height of the imaging lens system is **ImgH**, the following condition can be satisfied: $2.10 < f / \text{ImgH} < 3.80$. Therefore, it is favorable for providing proper viewing angles so that the imaging lens system is applicable to various applications.

(69) When a curvature radius of an object-side surface of the sixth lens element is **R11**, and a curvature radius of an image-side surface of the sixth lens element is **R12**, the following condition

can be satisfied: $-0.50 < (R11 + R12) / (R11 - R12) < 15.0$. Therefore, it is favorable for balancing the light path in both tangential direction and sagittal direction so as to correct astigmatism. Moreover, the following condition can also be satisfied: $-0.20 < (R11 + R12) / (R11 - R12) < 3.0$.

(70) When the maximum effective radius of the object-side surface of the first lens element is $Y11$, a maximum effective radius of an object-side surface of the fifth lens element is $Y51$, and the maximum effective radius of the image-side surface of the eighth lens element is $Y82$, the following condition can be satisfied: $0.70 < (Y82 - Y51) / (Y11 - Y51) < 2.50$. Therefore, it is favorable for arranging the outer diameters of the object side, the middle part and the image side of the imaging lens system so as to increase the symmetry of the imaging lens system, to improve image quality and to reduce sensitivity. Please refer to FIG. 31, which shows a schematic view of $Y11$, $Y51$ and $Y82$ according to the 1st embodiment of the present disclosure.

(71) When the entrance pupil diameter of the imaging lens system is EPD, and the maximum image height of the imaging lens system is $ImgH$, the following condition can be satisfied:

$1.20 < EPD / ImgH < 2.80$. Therefore, it is favorable for providing sufficient light on the image surface so as to meet the requirement of image brightness.

(72) When the Abbe number of the first lens element is $V1$, the Abbe number of the second lens element is $V2$, the Abbe number of the third lens element is $V3$, the Abbe number of the fourth lens element is $V4$, the Abbe number of the fifth lens element is $V5$, the Abbe number of the sixth lens element is $V6$, the Abbe number of the seventh lens element is $V7$, the Abbe number of the eighth lens element is $V8$, the Abbe number of the i-th lens element is V_i , a refractive index of the first lens element is $N1$, a refractive index of the second lens element is $N2$, a refractive index of the third lens element is $N3$, a refractive index of the fourth lens element is $N4$, a refractive index of the fifth lens element is $N5$, a refractive index of the sixth lens element is $N6$, a refractive index of the seventh lens element is $N7$, a refractive index of the eighth lens element is $N8$, a refractive index of the i-th lens element is N_i , and a minimum value of V_i / N_i is $(V_i / N_i)_{min}$, the following condition can be satisfied: $9.0 < (V_i / N_i)_{min} < 11.80$, wherein $i = 1, 2, 3, 4, 5, 6, 7, 8$. Therefore, it is favorable for the imaging lens system to have a sufficient image correction capability so as to correct various aberrations. Moreover, the following condition can also be satisfied:

$9.5 < (V_i / N_i)_{min} < 11.5$.

(73) When the axial distance between the object-side surface of the first lens element and the image surface is TL, the following condition can be satisfied: $4.0 \text{ [mm]} < TL < 8.0 \text{ [mm]}$. Therefore, it is favorable for controlling the total length of the imaging lens system, making the imaging lens system applicable to more applications so as to satisfy market demands. Moreover, the following condition can also be satisfied: $4.5 \text{ [mm]} < TL < 7.0 \text{ [mm]}$.

(74) When the maximum image height of the imaging lens system is $ImgH$, the following condition can be satisfied: $1.50 \text{ [mm]} < ImgH < 4.50 \text{ [mm]}$. Therefore, it is favorable for providing a proper light receiving area with sufficient image brightness so as to meet the requirement of product specifications. Moreover, the following condition can also be satisfied: $1.80 \text{ [mm]} < ImgH < 3.20 \text{ [mm]}$.

(75) According to the present disclosure, the aforementioned features and conditions can be utilized in numerous combinations so as to achieve corresponding effects.

(76) According to the present disclosure, the lens elements of the imaging lens system can be made of either glass or plastic material. When the lens elements are made of glass material, the refractive power distribution of the imaging lens system may be more flexible. The glass lens element can either be made by grinding or molding. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, which allows more control variables for eliminating aberrations thereof, the required number of the lens elements can be reduced, and the total track length of the imaging lens system can be effectively shortened. The aspheric surfaces may be formed by plastic injection molding or glass molding.

(77) According to the present disclosure, one or more of the lens elements' material may optionally include an additive which alters the lens elements' transmittance in a specific range of wavelength for a reduction in unwanted stray light or colour deviation. For example, the additive may optionally filter out light in the wavelength range of 600 nm to 800 nm to reduce excessive red light and/or near infrared light; or may optionally filter out light in the wavelength range of 350 nm to 450 nm to reduce excessive blue light and/or near ultraviolet light from interfering the final image. The additive may be homogeneously mixed with a plastic material to be used in manufacturing a mixed-material lens element by injection molding.

(78) According to the present disclosure, when a lens surface is aspheric, it means that the lens surface has an aspheric shape throughout its optically effective area, or a portion(s) thereof.

(79) According to the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly, unless otherwise stated, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof. Moreover, when a region of refractive power or focus of a lens element is not defined, it indicates that the region of refractive power or focus of the lens element is in the paraxial region thereof.

(80) According to the present disclosure, an inflection point is a point on the surface of the lens element at which the surface changes from concave to convex, or vice versa. A critical point is a non-axial point of the lens surface where its tangent is perpendicular to the optical axis.

(81) According to the present disclosure, an image surface of the imaging lens system, based on the corresponding image sensor, can be flat or curved, especially a curved surface being concave facing towards the object side of the imaging lens system.

(82) According to the present disclosure, an image correction unit, such as a field flattener, can be optionally disposed between the lens element closest to the image side of the imaging lens system and the image surface for correction of aberrations such as field curvature. The optical properties of the image correction unit, such as curvature, thickness, index of refraction, position and surface shape (convex or concave surface with spherical, aspheric, diffractive or Fresnel types), can be adjusted according to the design of an image capturing unit. In general, a preferable image correction unit is, for example, a thin transparent element having a concave object-side surface and a planar image-side surface, and the thin transparent element is disposed near the image surface.

(83) According to the present disclosure, the imaging lens system can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is set for eliminating the stray light and thereby improving image quality thereof.

(84) According to the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the imaging lens system and the image surface to produce a telecentric effect, and thereby improves the image-sensing efficiency of an image sensor (for example, CCD or CMOS). A middle stop disposed between the first lens element and the image surface is favorable for enlarging the viewing angle of the imaging lens system and thereby provides a wider field of view for the same.

(85) According to the present disclosure, the imaging lens system can include an aperture control unit. The aperture control unit may be a mechanical component or a light modulator, which can control the size and shape of the aperture through electricity or electrical signals. The mechanical component can include a movable member, such as a blade assembly or a light baffle. The light modulator can include a shielding element, such as a filter, an electrochromic material or a liquid-crystal layer. The aperture control unit controls the amount of incident light or exposure time to enhance the capability of image quality adjustment. In addition, the aperture control unit can be the

aperture stop of the present disclosure, which changes the f-number to obtain different image effects, such as the depth of field or lens speed.

(86) According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

(87) FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment. In FIG. 1, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor **199**. The imaging lens system includes, in order from an object side to an image side, a first lens element **110**, a second lens element **120**, an aperture stop **100**, a third lens element **130**, a fourth lens element **140**, a fifth lens element **150**, a sixth lens element **160**, a seventh lens element **170**, an eighth lens element **180**, a filter **190** and an image surface **195**. The imaging lens system includes eight lens elements (**110**, **120**, **130**, **140**, **150**, **160**, **170** and **180**) with no additional lens element disposed between each of the adjacent eight lens elements.

(88) The first lens element **110** with positive refractive power has an object-side surface **111** being convex in a paraxial region thereof and an image-side surface **112** being concave in a paraxial region thereof. The first lens element **110** is made of plastic material and has the object-side surface **111** and the image-side surface **112** being both aspheric.

(89) The second lens element **120** with positive refractive power has an object-side surface **121** being convex in a paraxial region thereof and an image-side surface **122** being convex in a paraxial region thereof. The second lens element **120** is made of plastic material and has the object-side surface **121** and the image-side surface **122** being both aspheric. The object-side surface **121** of the second lens element **120** has one inflection point. The image-side surface **122** of the second lens element **120** has two inflection points.

(90) The third lens element **130** with negative refractive power has an object-side surface **131** being convex in a paraxial region thereof and an image-side surface **132** being concave in a paraxial region thereof. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being both aspheric.

(91) The fourth lens element **140** with negative refractive power has an object-side surface **141** being convex in a paraxial region thereof and an image-side surface **142** being concave in a paraxial region thereof. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being both aspheric. The object-side surface **141** of the fourth lens element **140** has one inflection point. The image-side surface **142** of the fourth lens element **140** has one inflection point.

(92) The fifth lens element **150** with positive refractive power has an object-side surface **151** being convex in a paraxial region thereof and an image-side surface **152** being concave in a paraxial region thereof. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-side surface **152** being both aspheric. The object-side surface **151** of the fifth lens element **150** has one inflection point. The image-side surface **152** of the fifth lens element **150** has two inflection points.

(93) The sixth lens element **160** with positive refractive power has an object-side surface **161** being concave in a paraxial region thereof and an image-side surface **162** being convex in a paraxial region thereof. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being both aspheric. The object-side surface **161** of the sixth lens element **160** has two inflection points. The image-side surface **162** of the sixth lens element **160** has two inflection points.

(94) The seventh lens element **170** with negative refractive power has an object-side surface **171** being convex in a paraxial region thereof and an image-side surface **172** being concave in a

paraxial region thereof. The seventh lens element **170** is made of plastic material and has the object-side surface **171** and the image-side surface **172** being both aspheric. The object-side surface **171** of the seventh lens element **170** has three inflection points. The image-side surface **172** of the seventh lens element **170** has two inflection points. The image-side surface **172** of the seventh lens element **170** has at least one critical point in an off-axis region thereof.

(95) The eighth lens element **180** with negative refractive power has an object-side surface **181** being concave in a paraxial region thereof and an image-side surface **182** being convex in a paraxial region thereof. The eighth lens element **180** is made of plastic material and has the object-side surface **181** and the image-side surface **182** being both aspheric. The object-side surface **181** of the eighth lens element **180** has three inflection points. The image-side surface **182** of the eighth lens element **180** has one inflection point.

(96) The filter **190** is made of glass material and located between the eighth lens element **180** and the image surface **195**, and will not affect the focal length of the imaging lens system. The image sensor **199** is disposed on or near the image surface **195** of the imaging lens system.

(97) The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

(98) $X(Y) = (Y^2 / R) / (1 + \sqrt{1 - (1 + k) \times (Y / R)^2}) + \sum_i A_i \times (Y^i)$, where, X is the relative distance between a point on the aspheric surface spaced at a distance Y from an optical axis and the tangential plane at the aspheric surface vertex on the optical axis; Y is the vertical distance from the point on the aspheric surface to the optical axis; R is the curvature radius; k is the conic coefficient; and A_i is the i-th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14, 16 and 18.

(99) In the imaging lens system of the image capturing unit according to the 1st embodiment, when a focal length of the imaging lens system is f, an f-number of the imaging lens system is F_{no} , and half of a maximum field of view of the imaging lens system is HFOV, these parameters have the following values: f=5.72 millimeters (mm), F_{no} =1.55, HFOV=22.8 degrees (deg.).

(100) When an Abbe number of the eighth lens element **180** is V_8 , the following condition is satisfied: $V_8=18.7$.

(101) When a minimum value among Abbe numbers of all lens elements of the imaging lens system is V_{min} , the following condition is satisfied: $V_{min}=18.7$. In this embodiment, among the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180**, the Abbe number of the eighth lens element **180** is smaller than the Abbe numbers of the other lens elements, and V_{min} is equal to the Abbe number of the eighth lens element **180**.

(102) When an Abbe number of the first lens element **110** is V_1 , an Abbe number of the second lens element **120** is V_2 , an Abbe number of the third lens element **130** is V_3 , an Abbe number of the fourth lens element **140** is V_4 , an Abbe number of the fifth lens element **150** is V_5 , an Abbe number of the sixth lens element **160** is V_6 , an Abbe number of the seventh lens element **170** is V_7 , the Abbe number of the eighth lens element **180** is V_8 , and an Abbe number of the i-th lens element is V_i , the following condition is satisfied: $\sum V_i=270.0$, wherein $i=1, 2, 3, 4, 5, 6, 7, 8$.

(103) When a maximum value among refractive indices of all lens elements of the imaging lens system is N_{max} , the following condition is satisfied: $N_{max}=1.688$. In this embodiment, among the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180**, the refractive index of the eighth lens element **180** is larger than the refractive indices of the other lens elements, and N_{max} is equal to the refractive index of the eighth lens element **180**.

(104) When the Abbe number of the first lens element **110** is V_1 , the Abbe number of the second

lens element **120** is V_2 , the Abbe number of the third lens element **130** is V_3 , the Abbe number of the fourth lens element **140** is V_4 , the Abbe number of the fifth lens element **150** is V_5 , the Abbe number of the sixth lens element **160** is V_6 , the Abbe number of the seventh lens element **170** is V_7 , the Abbe number of the eighth lens element **180** is V_8 , the Abbe number of the i-th lens element is V_i , a refractive index of the first lens element **110** is N_1 , a refractive index of the second lens element **120** is N_2 , a refractive index of the third lens element **130** is N_3 , a refractive index of the fourth lens element **140** is N_4 , a refractive index of the fifth lens element **150** is N_5 , a refractive index of the sixth lens element **160** is N_6 , a refractive index of the seventh lens element **170** is N_7 , a refractive index of the eighth lens element **180** is N_8 , a refractive index of the i-th lens element is N_i , and a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, the following condition is satisfied:

$(V_i/N_i)_{\min}=11.08$. In this embodiment, $(V_i/N_i)_{\min}$ is equal to V_8/N_8 .

(105) When a sum of central thicknesses of all lens elements of the imaging lens system is ΣCT , and a sum of axial distances between each of all adjacent lens elements of the imaging lens system is ΣAT , the following condition is satisfied: $\Sigma CT/\Sigma AT=1.55$. In this embodiment, an axial distance between two adjacent lens elements is an air gap in a paraxial region between the two adjacent lens elements. In addition, in this embodiment, ΣCT is a sum of the central thicknesses of the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160**, the seventh lens element **170** and the eighth lens element **180**; ΣAT is a sum of the axial distance between the first lens element **110** and the second lens element **120**, the axial distance between the second lens element **120** and the third lens element **130**, the axial distance between the third lens element **130** and the fourth lens element **140**, the axial distance between the fourth lens element **140** and the fifth lens element **150**, the axial distance between the fifth lens element **150** and the sixth lens element **160**, the axial distance between the sixth lens element **160** and the seventh lens element **170**, and the axial distance between the seventh lens element **170** and the eighth lens element **180**.

(106) When a curvature radius of the object-side surface **111** of the first lens element **110** is R_1 , and a central thickness of the first lens element **110** is CT_1 , the following condition is satisfied: $R_1/CT_1=2.02$.

(107) When an axial distance between the image-side surface **182** of the eighth lens element **180** and the image surface **195** is BL , and the central thickness of the first lens element **110** is CT_1 , the following condition is satisfied: $BL/CT_1=0.37$.

(108) When a curvature radius of the image-side surface **142** of the fourth lens element **140** is R_8 , and the focal length of the imaging lens system is f , the following condition is satisfied: $R_8/f=0.47$.

(109) When a curvature radius of the object-side surface **161** of the sixth lens element **160** is R_{11} , and a curvature radius of the image-side surface **162** of the sixth lens element **160** is R_{12} , the following condition is satisfied: $(R_{11}+R_{12})/(R_{11}-R_{12})=2.23$.

(110) When a curvature radius of the object-side surface **171** of the seventh lens element **170** is R_{13} , and a curvature radius of the image-side surface **172** of the seventh lens element **170** is R_{14} , the following condition is satisfied: $(R_{13}-R_{14})/(R_{13}+R_{14})=0.13$.

(111) When the focal length of the imaging lens system is f , and a focal length of the first lens element **110** is f_1 , the following condition is satisfied: $f/f_1=0.94$.

(112) When the focal length of the imaging lens system is f , and a focal length of the fifth lens element **150** is f_5 , the following condition is satisfied: $f/f_5=0.17$.

(113) When the focal length of the fifth lens element **150** is f_5 , and a focal length of the seventh lens element **170** is f_7 , the following condition is satisfied: $f_5/f_7=-2.32$.

(114) When the focal length of the fifth lens element **150** is f_5 , and a focal length of the eighth lens element **180** is f_8 , the following condition is satisfied: $f_8/f_5=-0.17$.

(115) When the focal length of the imaging lens system is f , the focal length of the fifth lens element **150** is f_5 , and a focal length of the sixth lens element **160** is f_6 , the following condition is satisfied: $f/f_5+f/f_6=0.37$.

(116) When a maximum image height of the imaging lens system is ImgH , the following condition is satisfied: $\text{ImgH}=2.39$ [mm].

(117) When an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **195** is TL, the following condition is satisfied: $\text{TL}=5.54$ [mm].

(118) When the axial distance between the object-side surface **111** of the first lens element **110** and the image surface **195** is TL, and the focal length of the imaging lens system is f , the following condition is satisfied: $\text{TL}/f=0.97$.

(119) When the axial distance between the object-side surface **111** of the first lens element **110** and the image surface **195** is TL, and an entrance pupil diameter of the imaging lens system is EPD, the following condition is satisfied: $\text{TL}/\text{EPD}=1.50$.

(120) When an axial distance between the aperture stop **100** and the image surface **195** is SL, and the axial distance between the object-side surface **111** of the first lens element **110** and the image surface **195** is TL, the following condition is satisfied: $\text{SL}/\text{TL}=0.69$.

(121) When the axial distance between the image-side surface **182** of the eighth lens element **180** and the image surface **195** is BL, and an axial distance between the object-side surface **111** of the first lens element **110** and the image-side surface **182** of the eighth lens element **180** is TD, the following condition is satisfied: $\text{BL}/\text{TD}=0.07$.

(122) When the entrance pupil diameter of the imaging lens system is EPD, and the maximum image height of the imaging lens system is ImgH , the following condition is satisfied: $\text{EPD}/\text{ImgH}=1.54$.

(123) When the entrance pupil diameter of the imaging lens system is EPD, and the axial distance between the image-side surface **182** of the eighth lens element **180** and the image surface **195** is BL, the following condition is satisfied: $\text{EPD}/\text{BL}=10.32$.

(124) When the focal length of the imaging lens system is f , and the entrance pupil diameter of the imaging lens system is EPD, the following condition is satisfied: $f/\text{EPD}=1.55$.

(125) When a maximum effective radius of the object-side surface **111** of the first lens element **110** is $Y11$, the maximum image height of the imaging lens system is ImgH , and the entrance pupil diameter of the imaging lens system is EPD, the following condition is satisfied: $(Y11-\text{ImgH})/\text{EPD}=-0.12$.

(126) When the focal length of the imaging lens system is f , half of the maximum field of view of the imaging lens system is HFOV, and the entrance pupil diameter of the imaging lens system is EPD, the following condition is satisfied: $f \times \tan(\text{HFOV})/\text{EPD}=0.65$.

(127) When the maximum effective radius of the object-side surface **111** of the first lens element **110** is $Y11$, a maximum effective radius of the image-side surface **182** of the eighth lens element **180** is $Y82$, the axial distance between the image-side surface **182** of the eighth lens element **180** and the image surface **195** is BL, and the focal length of the imaging lens system is f , the following condition is satisfied: $(|Y11-Y82|+\text{BL})/f=0.11$.

(128) When the maximum effective radius of the object-side surface **111** of the first lens element **110** is $Y11$, a maximum effective radius of the object-side surface **151** of the fifth lens element **150** is $Y51$, and the maximum effective radius of the image-side surface **182** of the eighth lens element **180** is $Y82$, the following condition is satisfied: $(Y82-Y51)/(Y11-Y51)=1.35$.

(129) When the maximum effective radius of the object-side surface **111** of the first lens element **110** is $Y11$, the maximum effective radius of the image-side surface **182** of the eighth lens element **180** is $Y82$, the maximum image height of the imaging lens system is ImgH , and the focal length of the imaging lens system is f , the following condition is satisfied: $(|Y11-Y82|+|Y82-\text{ImgH}|)/f=0.08$.

(130) When the focal length of the imaging lens system is f , and the maximum image height of the imaging lens system is ImgH , the following condition is satisfied: $f/\text{ImgH}=2.39$.

(131) When a vertical distance between the at least one critical point on the image-side surface **172** of the seventh lens element **170** and the optical axis is $Yc72$, and the focal length of the imaging lens system is f , the following condition is satisfied: $Yc72/f=0.18$.

8.735 (ASP) 0.268 12 Lens 6 -31.526 (ASP) 0.339 Plastic 1.672 18.8 28.70 13 -12.017 (ASP) 0.476 14 Lens 7 2.034 (ASP) 0.195 Plastic 1.544 55.9 -14.85 15 1.570 (ASP) 0.450 16 Lens 8 -3.788 (ASP) 0.368 Plastic 1.688 18.7 -5.89 17 -60.027 (ASP) 0.200 18 Filter Plano 0.110 Glass 1.517 64.2 — 19 Plano 0.048 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line). (150) TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface # 1 2 3 4 6 7 k= -2.9818E-01 -2.1159E-01 1.9516E-01 9.0000E+01 -1.5515E+01 -7.7859E+00 A4= -1.3398E-04 1.9575E-02 2.2307E-02 -7.1350E-03 -1.0608E-01 -7.4307E-02 A6= 6.7041E-03 -4.2606E-02 -4.6342E-02 2.2191E-02 1.8304E-01 2.4706E-01 A8= -6.8806E-03 3.5360E-02 3.7358E-02 -1.7351E-02 -1.7393E-01 -2.6206E-01 A10= 3.5068E-03 -1.4054E-02 -1.4188E-02 7.1729E-03 9.9062E-02 1.6566E-01 A12= -8.1799E-04 2.6735E-03 2.5169E-03 -1.6202E-03 -2.8945E-02 -5.5348E-02 A14= 6.4288E-05 -1.9300E-04 -1.7231E-04 1.5350E-04 3.4512E-03 1.2167E-02 Surface # 8 9 10 11 12 13 k= -6.1750E+00 1.5329E+00 -2.6403E+01 -9.4358E+00 4.7871E+01 2.2295E+01 A4= -3.8548E-02 -9.4467E-02 -6.2958E-02 -6.0453E-02 -1.7510E-02 -5.4373E-02 A6= 1.2033E-01 1.5860E-01 -1.8085E-01 -6.7026E-02 -5.7394E-02 1.6055E-02 A8= -1.1681E-01 -1.9405E-01 3.5816E-01 1.4773E-01 1.5074E-01 1.1166E-02 A10= 2.1499E-02 7.1861E-03 -5.6492E-01 -1.8999E-01 -1.0806E-01 1.1721E-03 A12= 3.3097E-02 1.4223E-01 4.7251E-01 1.5808E-01 3.6239E-02 -3.4809E-03 A14= -1.3689E-02 -1.0679E-01 -2.0462E-01 -6.3406E-02 -5.6969E-03 9.5901E-04 A16= — 2.4002E-02 3.6694E-02 9.3902E-03 2.9652E-04 -8.8434E-05 Surface # 14 15 16 17 k= -6.6696E+00 -1.9700E+00 -6.2564E+01 9.0000E+01 A4= -2.0747E-01 -2.4561E-01 -3.3748E-01 -3.7607E-01 A6= 1.5432E-02 1.1142E-01 5.4640E-01 4.3917E-01 A8= -4.1439E-02 -2.7705E-02 -4.1852E-01 -2.6820E-01 A10= 8.1875E-02 -1.3161E-03 1.8359E-01 9.4698E-02 A12= -5.0008E-02 1.4485E-03 -4.8465E-02 -1.9049E-02 A14= 1.4026E-02 3.8338E-04 7.5566E-03 1.9194E-03 A16= -1.7700E-03 -2.4250E-04 -6.3422E-04 -5.1294E-05 A18= 7.1084E-05 2.8498E-05 2.1805E-05 -3.3209E-06

(151) In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-20 represent the surfaces sequentially arranged from the object side to the image side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-18 represent the aspheric coefficients ranging from the 4th order to the 18th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

(152) FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment. In FIG. 3, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 299. The imaging lens system includes, in order from an object side to an image side, a first lens element 210, a second lens element 220, an aperture stop 200, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, a seventh lens element 270, an eighth lens element 280, a filter 290 and an image surface 295. The imaging lens system includes eight lens elements (210, 220, 230, 240, 250, 260, 270 and 280) with no additional lens element disposed between each of the adjacent eight lens elements.

(153) The first lens element 210 with positive refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being convex in a paraxial region thereof. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being both aspheric. The object-side surface 211 of the first

lens element **210** has one inflection point. The image-side surface **212** of the first lens element **210** has three inflection points.

(154) The second lens element **220** with negative refractive power has an object-side surface **221** being concave in a paraxial region thereof and an image-side surface **222** being convex in a paraxial region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric. The object-side surface **221** of the second lens element **220** has one inflection point.

(155) The third lens element **230** with negative refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

(156) The fourth lens element **240** with negative refractive power has an object-side surface **241** being convex in a paraxial region thereof and an image-side surface **242** being concave in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric. The object-side surface **241** of the fourth lens element **240** has one inflection point. The image-side surface **242** of the fourth lens element **240** has one inflection point.

(157) The fifth lens element **250** with positive refractive power has an object-side surface **251** being convex in a paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric. The object-side surface **251** of the fifth lens element **250** has two inflection points. The image-side surface **252** of the fifth lens element **250** has two inflection points.

(158) The sixth lens element **260** with positive refractive power has an object-side surface **261** being concave in a paraxial region thereof and an image-side surface **262** being convex in a paraxial region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric. The object-side surface **261** of the sixth lens element **260** has two inflection points. The image-side surface **262** of the sixth lens element **260** has two inflection points.

(159) The seventh lens element **270** with positive refractive power has an object-side surface **271** being convex in a paraxial region thereof and an image-side surface **272** being concave in a paraxial region thereof. The seventh lens element **270** is made of plastic material and has the object-side surface **271** and the image-side surface **272** being both aspheric. The object-side surface **271** of the seventh lens element **270** has two inflection points. The image-side surface **272** of the seventh lens element **270** has two inflection points. The image-side surface **272** of the seventh lens element **270** has at least one critical point in an off-axis region thereof.

(160) The eighth lens element **280** with negative refractive power has an object-side surface **281** being concave in a paraxial region thereof and an image-side surface **282** being convex in a paraxial region thereof. The eighth lens element **280** is made of plastic material and has the object-side surface **281** and the image-side surface **282** being both aspheric. The object-side surface **281** of the eighth lens element **280** has one inflection point. The image-side surface **282** of the eighth lens element **280** has one inflection point.

(161) The filter **290** is made of glass material and located between the eighth lens element **280** and the image surface **295**, and will not affect the focal length of the imaging lens system. The image sensor **299** is disposed on or near the image surface **295** of the imaging lens system.

(162) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **280** is V_8 , and a refractive index of the eighth lens element **280** is N_8 .

(163) The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric

surface data are shown in Table 4 below.

(164) TABLE-US-00003 TABLE 3 2nd Embodiment $f = 5.13$ mm, $Fno = 1.56$, $HFOV = 23.3$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Lens 1 2.010 (ASP) 0.980 Plastic 1.545 56.1 3.22 2 -11.396 (ASP) 0.077 3 Lens 2 -11.113
(ASP) 0.255 Plastic 1.650 28.0 -58.50 4 -15.844 (ASP) -0.066 5 Ape. Stop Plano 0.092 6 Lens 3
4.642 (ASP) 0.180 Plastic 1.669 19.5 -8.51 7 2.517 (ASP) 0.242 8 Lens 4 4.751 (ASP) 0.278
Plastic 1.639 23.5 -9.60 9 2.616 (ASP) 0.522 10 Lens 5 5.448 (ASP) 0.283 Plastic 1.650 21.5
13.02 11 14.961 (ASP) 0.200 12 Lens 6 -18.166 (ASP) 0.574 Plastic 1.672 18.8 114.95 13 -14.894
(ASP) 0.629 14 Lens 7 1.704 (ASP) 0.292 Plastic 1.544 55.9 367.07 15 1.615 (ASP) 0.666 16 Lens
8 -2.878 (ASP) 0.279 Plastic 1.688 18.7 -5.02 17 -17.953 (ASP) 0.200 18 Filter Plano 0.110
Glass 1.517 64.2 — 19 Plano 0.091 20 Image Plano — Note: Reference wavelength is 587.6 nm
(d-line).

(165) TABLE-US-00004 TABLE 4 Aspheric Coefficients Surface # 1 2 3 4 6 7 $k = -3.8546E-01$
 $-5.0000E+01$ $-3.8697E+01$ $6.3869E+01$ $-5.5442E+00$ $-8.6951E+00$ $A4 = -2.4107E-03$
 $1.1382E-02$ $-2.4848E-02$ $2.7928E-02$ $-6.9479E-02$ $-1.3821E-01$ $A6 = 1.8521E-02$
 $5.8912E-03$ $4.9651E-02$ $-1.6609E-02$ $2.3298E-01$ $3.5136E-01$ $A8 = -2.0649E-02$
 $-6.4475E-03$ $-3.4581E-02$ $-3.9789E-03$ $-3.3170E-01$ $-2.0806E-01$ $A10 = 1.2040E-02$
 $7.8321E-04$ $9.9460E-03$ $6.4554E-03$ $2.2356E-01$ $-6.7145E-02$ $A12 = -3.4675E-03$
 $2.0852E-04$ $-8.8650E-04$ $-2.0111E-03$ $-6.9416E-02$ $1.2180E-01$ $A14 = 3.2888E-04$ — —
 $1.8438E-04$ $8.1147E-03$ $-3.3562E-02$ Surface # 8 9 10 11 12 13 $k = -5.7395E+00$
 $1.8708E+00$ $-1.3183E+01$ $-9.0000E+01$ $1.2407E+01$ $3.5550E+01$ $A4 = -2.1465E-01$
 $-1.7568E-01$ $-2.0134E-01$ $-1.7781E-01$ $-9.0865E-02$ $-9.2370E-02$ $A6 = 4.8879E-01$
 $2.9066E-01$ $3.1565E-01$ $2.4726E-01$ $1.5525E-01$ $1.0693E-01$ $A8 = -3.4607E-01$
 $-6.2755E-02$ $-6.8233E-01$ $-3.3296E-01$ $-8.7085E-02$ $-6.6080E-02$ $A10 = 3.7788E-02$
 $-3.5340E-01$ $9.0729E-01$ $3.6526E-01$ $2.7898E-02$ $2.8529E-02$ $A12 = 6.1480E-02$
 $4.2034E-01$ $-7.4012E-01$ $-2.8247E-01$ $-5.1207E-03$ $-5.9526E-03$ $A14 = -2.0640E-02$
 $-2.0387E-01$ $3.1004E-01$ $1.2454E-01$ $3.9186E-04$ $1.6928E-04$ $A16 = 3.6974E-02$
 $-4.9699E-02$ $-2.1852E-02$ — $6.4342E-05$ Surface # 14 15 16 17 $k = -4.9523E+00$
 $-1.7479E+00$ $-1.0364E+01$ $5.9343E+01$ $A4 = -1.5260E-01$ $-2.1352E-01$ $-2.8107E-01$
 $-3.7829E-01$ $A6 = 6.0657E-03$ $4.7865E-02$ $3.8726E-01$ $4.8242E-01$ $A8 = 3.6583E-02$
 $3.3178E-02$ $-2.3914E-01$ $-2.9209E-01$ $A10 = -2.2972E-02$ $-3.3136E-02$ $8.2801E-02$
 $1.0396E-01$ $A12 = 6.9555E-03$ $1.2058E-02$ $-1.6657E-02$ $-2.2821E-02$ $A14 =$
 $-1.0604E-03$ $-2.1120E-03$ $1.8498E-03$ $3.0493E-03$ $A16 = 6.4788E-05$ $1.4854E-04$
 $-9.3831E-05$ $-2.2789E-04$ $A18 =$ — — $9.5301E-07$ $7.3197E-06$

(166) In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

(167) Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

(168) TABLE-US-00005 2nd Embodiment f [mm] 5.13 EPD/BL 8.20 Fno 1.56 f/EPD 1.56 $HFOV$ [deg.] 23.3 $(Y11 - ImgH)/EPD$ -0.20 $V8$ 18.7 $f \times \tan(HFOV)/EPD$ 0.67 $Vmin$ 18.7 $(|Y11 - Y82| + BL)/f$ 0.17 ΣVi 241.9 $(Y82 - Y51)/(Y11 - Y51)$ 1.96 $Nmax$ 1.688 $(|Y11 - Y82| + |Y82 - ImgH|)/f$ 0.13 $(Vi/Ni)min$ 11.08 $f/ImgH$ 2.15 $\Sigma CT/\Sigma AT$ 1.32 $Yc72/f$ 0.20 $R1/CT1$ 2.05 $|f/R1| + |f/R2|$ 3.00 $BL/CT1$ 0.41 $|f/R3| + |f/R4|$ 0.79 $R8/f$ 0.51 $|f/R5| + |f/R6|$ 3.14 $(R11 + R12)/10.10$ $|f/R7| + |f/R8|$ 3.04 $(R11 - R12)(R13 - R14)/0.03$ $|f/R9| + |f/R10|$ 1.28 $(R13 + R14)f/f1$ 1.59 $|f/R11| + |f/R12|$ 0.63 $f/f5$ 0.39 $|f/R13| + |f/R14|$ 6.19 $f5/f7$ 0.04 $|f/R15| + |f/R16|$ 2.07 $f8/f5$ -0.39 $V1/N1$ 36.30 $f/f5 + f/f6$ 0.44 $V2/N2$ 16.97 $ImgH$ [mm] 2.39 $V3/N3$ 11.66 TL [mm] 5.88 $V4/N4$ 14.34 TL/f 1.15 $V5/N5$ 13.01 TL/EPD 1.79 $V6/N6$ 11.24 SL/TL 0.79 $V7/N7$ 36.23 BL/TD 0.07 $V8/N8$ 11.08 $EPD/ImgH$

3rd Embodiment

(169) FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment. In FIG. 5, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 399. The imaging lens system includes, in order from an object side to an image side, a first lens element 310, a second lens element 320, an aperture stop 300, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, a seventh lens element 370, an eighth lens element 380, a filter 390 and an image surface 395. The imaging lens system includes eight lens elements (310, 320, 330, 340, 350, 360, 370 and 380) with no additional lens element disposed between each of the adjacent eight lens elements.

(170) The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being concave in a paraxial region thereof. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

(171) The second lens element 320 with positive refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being convex in a paraxial region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric. The object-side surface 321 of the second lens element 320 has one inflection point. The image-side surface 322 of the second lens element 320 has two inflection points.

(172) The third lens element 330 with negative refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric.

(173) The fourth lens element 340 with negative refractive power has an object-side surface 341 being convex in a paraxial region thereof and an image-side surface 342 being concave in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric. The image-side surface 342 of the fourth lens element 340 has one inflection point.

(174) The fifth lens element 350 with positive refractive power has an object-side surface 351 being convex in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric. The object-side surface 351 of the fifth lens element 350 has two inflection points. The image-side surface 352 of the fifth lens element 350 has two inflection points.

(175) The sixth lens element 360 with positive refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being planar in a paraxial region thereof. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being both aspheric. The object-side surface 361 of the sixth lens element 360 has three inflection points. The image-side surface 362 of the sixth lens element 360 has three inflection points.

(176) The seventh lens element 370 with negative refractive power has an object-side surface 371 being convex in a paraxial region thereof and an image-side surface 372 being concave in a paraxial region thereof. The seventh lens element 370 is made of plastic material and has the object-side surface 371 and the image-side surface 372 being both aspheric. The object-side surface 371 of the seventh lens element 370 has three inflection points. The image-side surface 372 of the seventh lens element 370 has two inflection points. The image-side surface 372 of the seventh lens

element **370** has at least one critical point in an off-axis region thereof.

(177) The eighth lens element **380** with negative refractive power has an object-side surface **381** being concave in a paraxial region thereof and an image-side surface **382** being convex in a paraxial region thereof. The eighth lens element **380** is made of plastic material and has the object-side surface **381** and the image-side surface **382** being both aspheric. The object-side surface **381** of the eighth lens element **380** has four inflection points. The image-side surface **382** of the eighth lens element **380** has four inflection points.

(178) The filter **390** is made of glass material and located between the eighth lens element **380** and the image surface **395**, and will not affect the focal length of the imaging lens system. The image sensor **399** is disposed on or near the image surface **395** of the imaging lens system.

(179) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **380** is V_8 , and a refractive index of the eighth lens element **380** is N_8 .

(180) The focal length of the imaging lens system is f , a curvature radius of an object-side surface of one lens element of the imaging lens system is R_o , a curvature radius of an image-side surface of the lens element is R_i , and one (the sixth lens element **360**) of the eight lens elements (**310**, **320**, **330**, **340**, **350**, **360**, **370** and **380**) satisfies $|f/R_o|+|f/R_i|<0.50$. In detail, a curvature radius of the object-side surface **361** of the sixth lens element **360** is R_{11} , a curvature radius of the image-side surface **362** of the sixth lens element **360** is R_{12} , and $|f/R_{11}|+|f/R_{12}|=0.26$.

(181) The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

(182) TABLE-US-00006 TABLE 5 3rd Embodiment $f = 5.44$ mm, $Fno = 1.55$, $HFOV = 23.4$ deg.

Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	0 Object	Plano	Infinity
1 Lens 1	1.933 (ASP)	1.000	Plastic	1.545	56.1	5.24	2	4.882 (ASP)	0.034
3 Lens 2	4.368 (ASP)	0.525	Plastic	1.545	56.1	6.85	4	-24.642 (ASP)	0.037
5 Ape. Stop	Plano	0.046	6 Lens 3	5.938 (ASP)	0.160	Plastic	1.669	19.5	-7.00
7	2.589 (ASP)	0.241	8 Lens 4	5.106 (ASP)	0.278	Plastic	1.639	23.5	-10.42
9	2.828 (ASP)	0.477	10 Lens 5	5.698 (ASP)	0.168	Plastic	1.650	21.5	27.93
11	8.205 (ASP)	0.181	12 Lens 6	21.021 (ASP)	0.351	Plastic	1.672	18.8	31.29
13	∞ (ASP)	0.600	14 Lens 7	1.557 (ASP)	0.259	Plastic	1.544	55.9	-19.57
15	1.279 (ASP)	0.453	16 Lens 8	-3.923 (ASP)	0.327	Plastic	1.688	18.7	-6.98
17	-22.218 (ASP)	0.200	18 Filter	Plano	0.110	Glass	1.517	64.2	—
19	Plano	0.096	20 Image	Plano	—	Note: Reference wavelength is 587.6 nm (d-line).			

(183) TABLE-US-00007 TABLE 6 Aspheric Coefficients

Surface #	1	2	3	4	6	7	k=			
1	-2.8473E-01	-2.0699E-01	7.1295E-02	8.2073E+01	-1.3421E+01	-7.0746E+00	A4= 1.8820E-03			
2	-1.9981E-03	-2.4268E-03	1.2180E-02	-4.1338E-02	1.6480E-03	A6= 1.0394E-03				
3	4.2295E-03	6.4874E-03	-9.3393E-04	6.6041E-03	1.4507E-03	A8= -5.6986E-04				
4	-2.5212E-03	-3.2498E-03	-1.7021E-03	3.3731E-02	3.5160E-02	A10= 2.8654E-04				
6	4.5154E-04	3.6115E-04	6.7222E-04	-2.5193E-02	-3.0206E-04	A12= -6.1949E-05	—			
7	-1.7790E-04	7.9099E-03	-5.4998E-03	A14= —	—	2.3254E-05	-9.8203E-04			
8	2.8390E-03	Surface # 8	9	10	11	12	13	k= -4.9629E+00	1.7708E+00	-1.9480E+01
9	-8.4012E+01	-5.0000E+01	3.6991E+01	A4= -1.5027E-02	-6.6964E-02	-4.2935E-02				
10	-7.0940E-02	-9.5163E-02	-1.2368E-01	A6= -1.7762E-03	3.1895E-02	-2.9514E-01				
11	1.3155E-02	1.6925E-01	1.8191E-01	A8= 3.5862E-02	-4.3379E-02	6.1495E-01				
12	-1.2811E-01	-9.8991E-02	-1.4316E-01	A10= -2.3655E-02	4.1881E-02	-9.7046E-01				
13	2.8721E-01	2.9907E-02	7.4071E-02	A12= 1.2553E-02	-2.7510E-02	9.4500E-01				
14	-2.6435E-01	-4.1308E-03	-2.1950E-02	A14= -3.0673E-03	4.9500E-03	-5.0473E-01				
15	1.1535E-01	9.8804E-05	3.3473E-03	A16= —	-2.4421E-04	1.0934E-01	-1.9269E-02	—		
16	-2.1886E-04	Surface # 14	15	16	17	k= -6.2900E+00	-1.8764E+00	-3.7911E+01	9.0000E+01	
17	A4= -1.5798E-01	-2.6035E-01	-3.7070E-01	-4.8686E-01	A6= -5.8481E-02	1.2129E-01				
18	6.2273E-01	6.4468E-01	A8= 8.0356E-02	-2.6659E-02	-4.9667E-01	-4.3100E-01	A10=			

-2.1259E-02 -3.5085E-03 2.2892E-01 1.6724E-01 A12= -4.2909E-03 3.5173E-03
-6.4120E-02 -3.8509E-02 A14= 2.9734E-03 -8.9257E-04 1.0694E-02 5.0442E-03
A16= -3.8440E-04 9.6523E-05 -9.6727E-04 -3.2801E-04 A18= -2.4892E-06
3.6172E-05 7.0598E-06

(184) In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

(185) Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

(186) TABLE-US-00008 3rd Embodiment f [mm] 5.44 EPD/BL 8.65 Fno 1.55 f/EPD 1.55 HFOV [deg.] 23.4 (Y11 - ImgH)/EPD -0.15 V8 18.7 $f \times \tan(\text{HFOV})/\text{EPD}$ 0.67 Vmin 18.7 (|Y11 - Y82| + BL)/f 0.13 ΣV_i 270.0 (Y82 - Y51)/(Y11 - Y51) 1.49 Nmax 1.688 (|Y11 - Y82| + |Y82 - ImgH|)/f 0.09 (Vi/Ni)min 11.08 f/ImgH 2.27 $\Sigma \text{CT}/\Sigma \text{AT}$ 1.48 Yc72/f 0.21 R1/CT1 1.93 |f/R1| + |f/R2| 3.93 BL/CT1 0.41 |f/R3| + |f/R4| 1.47 R8/f 0.52 |f/R5| + |f/R6| 3.01 (R11 + R12)/-1.00 |f/R7| + |f/R8| 2.99 (R11 - R12) (R13 - R14)/ 0.10 |f/R9| + |f/R10| 1.62 (R13 + R14) f/f1 1.04 |f/R11| + |f/R12| 0.26 f/f5 0.19 |f/R13| + |f/R14| 7.74 f5/f7 -1.43 |f/R15| + |f/R16| 1.63 f8/f5 -0.25 V1/N1 36.30 f/f5 + f/f6 0.37 V2/N2 36.30 ImgH [mm] 2.39 V3/N3 11.66 TL [mm] 5.54 V4/N4 14.34 TL/f 1.02 V5/N5 13.01 TL/EPD 1.58 V6/N6 11.24 SL/TL 0.71 V7/N7 36.23 BL/TD 0.08 V8/N8 11.08 EPD/ImgH 1.47 — —

4th Embodiment

(187) FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment. In FIG. 7, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 499. The imaging lens system includes, in order from an object side to an image side, a first lens element 410, an aperture stop 400, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, a seventh lens element 470, an eighth lens element 480, a filter 490 and an image surface 495. The imaging lens system includes eight lens elements (410, 420, 430, 440, 450, 460, 470 and 480) with no additional lens element disposed between each of the adjacent eight lens elements.

(188) The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric. The image-side surface 412 of the first lens element 410 has one inflection point.

(189) The second lens element 420 with positive refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric. The object-side surface 421 of the second lens element 420 has one inflection point. The image-side surface 422 of the second lens element 420 has one inflection point.

(190) The third lens element 430 with negative refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric.

(191) The fourth lens element 440 with negative refractive power has an object-side surface 441 being convex in a paraxial region thereof and an image-side surface 442 being concave in a

paraxial region thereof. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric. The image-side surface **442** of the fourth lens element **440** has one inflection point.

(192) The fifth lens element **450** with positive refractive power has an object-side surface **451** being convex in a paraxial region thereof and an image-side surface **452** being concave in a paraxial region thereof. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric. The object-side surface **451** of the fifth lens element **450** has one inflection point. The image-side surface **452** of the fifth lens element **450** has three inflection points.

(193) The sixth lens element **460** with negative refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof. The sixth lens element **460** is made of plastic material and has the object-side surface **461** and the image-side surface **462** being both aspheric. The object-side surface **461** of the sixth lens element **460** has three inflection points. The image-side surface **462** of the sixth lens element **460** has three inflection points.

(194) The seventh lens element **470** with positive refractive power has an object-side surface **471** being convex in a paraxial region thereof and an image-side surface **472** being concave in a paraxial region thereof. The seventh lens element **470** is made of plastic material and has the object-side surface **471** and the image-side surface **472** being both aspheric. The object-side surface **471** of the seventh lens element **470** has three inflection points. The image-side surface **472** of the seventh lens element **470** has two inflection points. The image-side surface **472** of the seventh lens element **470** has at least one critical point in an off-axis region thereof.

(195) The eighth lens element **480** with negative refractive power has an object-side surface **481** being concave in a paraxial region thereof and an image-side surface **482** being convex in a paraxial region thereof. The eighth lens element **480** is made of plastic material and has the object-side surface **481** and the image-side surface **482** being both aspheric. The object-side surface **481** of the eighth lens element **480** has four inflection points. The image-side surface **482** of the eighth lens element **480** has four inflection points.

(196) The filter **490** is made of glass material and located between the eighth lens element **480** and the image surface **495**, and will not affect the focal length of the imaging lens system. The image sensor **499** is disposed on or near the image surface **495** of the imaging lens system.

(197) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **480** is V_8 , and a refractive index of the eighth lens element **480** is N_8 .

(198) The focal length of the imaging lens system is f , a curvature radius of an object-side surface of one lens element of the imaging lens system is R_o , a curvature radius of an image-side surface of the lens element is R_i , and one (the sixth lens element **460**) of the eight lens elements (**410**, **420**, **430**, **440**, **450**, **460**, **470** and **480**) satisfies $|f/R_o| + |f/R_i| < 0.50$. In detail, a curvature radius of the object-side surface **461** of the sixth lens element **460** is R_{11} , a curvature radius of the image-side surface **462** of the sixth lens element **460** is R_{12} , and $|f/R_{11}| + |f/R_{12}| = 0.09$.

(199) The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

(200) TABLE-US-00009 TABLE 7 4th Embodiment $f = 5.11$ mm, $Fno = 1.57$, $HFOV = 24.2$ deg.

Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	0 Object	Plano	Infinity
1 Lens 1	2.020 (ASP)	0.914	Plastic	1.545	56.1	5.80	2	4.710 (ASP)	0.537
3 Ape. Stop	Plano								
4 Lens 2	3.010 (ASP)	0.441	Plastic	1.545	56.1	7.01	5	13.493 (ASP)	0.033
6 Lens 3	4.232 (ASP)	0.170	Plastic	1.669	19.5	-12.86	7	2.790 (ASP)	0.224
8 Lens 4	5.971 (ASP)	0.288	Plastic	1.639	23.5	-7.33	9	2.575 (ASP)	0.498
10 Lens 5	5.170 (ASP)	0.233	Plastic	1.634	23.8	16.48	11	10.052 (ASP)	0.213
12 Lens 6	248.862 (ASP)	0.394	Plastic	1.672	18.8	-147.01	13	70.673 (ASP)	

0.358 14 Lens 7 1.376 (ASP) 0.332 Plastic 1.544 55.9 55.58 15 1.319 (ASP) 0.439 16 Lens 8
 -3.198 (ASP) 0.493 Plastic 1.688 18.7 -5.05 17 -42.623 (ASP) 0.200 18 Filter Plano 0.110 Glass
 1.517 64.2 — 19 Plano 0.093 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).
 (201) TABLE-US-00010 TABLE 8 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k = -3.4968E-01$
 $-2.0956E+00$ $5.0721E-01$ $-8.8110E+01$ $-1.3202E+01$ $-7.1241E+00$ $A4 = 1.1721E-03$
 $-1.2972E-03$ $5.4549E-03$ $-3.5948E-02$ $-1.2265E-01$ $-4.7252E-02$ $A6 = 7.3647E-04$
 $-4.1381E-03$ $-1.1228E-02$ $1.2482E-01$ $3.1370E-01$ $2.1096E-01$ $A8 = 1.0042E-03$
 $2.1185E-03$ $1.1629E-02$ $-1.3875E-01$ $-3.9570E-01$ $-2.6860E-01$ $A10 = -1.1053E-03$
 $-3.2130E-04$ $-3.4987E-03$ $7.6075E-02$ $2.6774E-01$ $1.8015E-01$ $A12 = 4.0452E-04$ — —
 $-2.1630E-02$ $-9.2147E-02$ $-3.7392E-02$ $A14 = -4.9171E-05$ — — $2.5688E-03$
 $1.2722E-02$ $-1.4644E-03$ Surface # 8 9 10 11 12 13 $k = -8.2316E+00$ $1.5801E+00$
 $1.5540E+00$ $-6.5852E+01$ $-9.0000E+01$ $-8.1932E+01$ $A4 = -4.0957E-02$ $-4.9406E-02$
 $-6.4130E-02$ $-8.4105E-02$ $-9.7195E-02$ $-1.8836E-01$ $A6 = 1.2456E-01$ $8.3163E-03$
 $-2.2198E-01$ $-5.2042E-02$ $1.7255E-01$ $3.0782E-01$ $A8 = -2.2457E-01$ $-2.0170E-02$
 $5.8295E-01$ $1.7177E-01$ $-9.5845E-02$ $-2.5004E-01$ $A10 = 2.3105E-01$ $-1.2852E-02$
 $-9.5006E-01$ $-1.9332E-01$ $2.6121E-02$ $1.2825E-01$ $A12 = -1.1076E-01$ $4.5849E-02$
 $8.3052E-01$ $1.0183E-01$ $-3.1740E-03$ $-4.0188E-02$ $A14 = 2.1589E-02$ $-3.9534E-02$
 $-3.7594E-01$ $-1.9452E-02$ $5.2188E-05$ $6.9832E-03$ $A16 =$ — $1.0442E-02$ $6.8047E-02$
 $-2.4035E-05$ — $-5.2202E-04$ Surface # 14 15 16 17 $k = -5.2821E+00$ $-1.7111E+00$
 $-4.5299E+01$ $7.0253E+01$ $A4 = -2.3302E-01$ $-3.1245E-01$ $-3.7587E-01$ $-4.8200E-01$ $A6 =$
 $1.6306E-01$ $2.8410E-01$ $6.8210E-01$ $7.0440E-01$ $A8 = -1.4779E-01$ $-2.2614E-01$
 $-6.3365E-01$ $-5.7091E-01$ $A10 = 1.0030E-01$ $1.2843E-01$ $3.6597E-01$ $2.9805E-01$ $A12 =$
 $-3.9799E-02$ $-4.8245E-02$ $-1.3717E-01$ $-1.0387E-01$ $A14 = 8.3399E-03$ $1.1107E-02$
 $3.3029E-02$ $2.4110E-02$ $A16 = -7.1097E-04$ $-1.4261E-03$ $-4.8998E-03$ $-3.5971E-03$ $A18 =$
 — $7.9558E-05$ $4.0727E-04$ $3.1337E-04$ $A20 =$ — — $-1.4552E-05$ $-1.2129E-05$

(202) In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

(203) Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

(204) TABLE-US-00011 4th Embodiment f [mm] 5.11 EPD/BL 8.06 F_{no} 1.57 f /EPD 1.57 HFOV [deg.] 24.2 $(Y_{11} - \text{ImgH})/\text{EPD}$ -0.18 V_8 18.7 $f \times \tan(\text{HFOV})/\text{EPD}$ 0.70 V_{min} 18.7 $(|Y_{11} - Y_{82}| + \text{BL})/f$ 0.16 ΣV_i 272.4 $(Y_{82} - Y_{51})/(Y_{11} - Y_{51})$ 1.71 N_{max} 1.688 $(|Y_{11} - Y_{82}| + |Y_{82} - \text{ImgH}|)/f$ 0.12 $(V_i/N_i)_{min}$ 11.08 f/ImgH 2.14 $\Sigma CT/\Sigma AT$ 1.64 Y_{c72}/f 0.23 R_1/CT_1 2.21 $|f/R_1| + |f/R_2|$ 3.61 BL/CT_1 0.44 $|f/R_3| + |f/R_4|$ 2.07 R_8/f 0.50 $|f/R_5| + |f/R_6|$ 3.04 $(R_{11} + R_{12})/$ 1.79 $|f/R_7| + |f/R_8|$ 2.84 $(R_{11} - R_{12})$ $(R_{13} - R_{14})/$ 0.02 $|f/R_9| + |f/R_{10}|$ 1.50 $(R_{13} + R_{14})$ f/f_1 0.88 $|f/R_{11}| + |f/R_{12}|$ 0.09 f/f_5 0.31 $|f/R_{13}| + |f/R_{14}|$ 7.58 f_5/f_7 0.30 $|f/R_{15}| + |f/R_{16}|$ 1.72 f_8/f_5 -0.31 V_1/N_1 36.30 $f/f_5 + f/f_6$ 0.28 V_2/N_2 36.30 ImgH [mm] 2.39 V_3/N_3 11.66 TL [mm] 5.66 V_4/N_4 14.34 TL/f 1.11 V_5/N_5 14.59 TL/EPD 1.74 V_6/N_6 11.24 SL/TL 0.74 V_7/N_7 36.23 BL/TD 0.08 V_8/N_8 11.08 EPD/ImgH 1.36 — —

5th Embodiment

(205) FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment. In FIG. 9, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 599. The imaging lens system includes, in order from an object side to an image side, a first lens element 510, an aperture stop 500, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens

element **550**, a sixth lens element **560**, a seventh lens element **570**, an eighth lens element **580**, a filter **590** and an image surface **595**. The imaging lens system includes eight lens elements (**510**, **520**, **530**, **540**, **550**, **560**, **570** and **580**) with no additional lens element disposed between each of the adjacent eight lens elements.

(206) The first lens element **510** with positive refractive power has an object-side surface **511** being convex in a paraxial region thereof and an image-side surface **512** being concave in a paraxial region thereof. The first lens element **510** is made of plastic material and has the object-side surface **511** and the image-side surface **512** being both aspheric

(207) The second lens element **520** with positive refractive power has an object-side surface **521** being convex in a paraxial region thereof and an image-side surface **522** being concave in a paraxial region thereof. The second lens element **520** is made of plastic material and has the object-side surface **521** and the image-side surface **522** being both aspheric. The object-side surface **521** of the second lens element **520** has one inflection point. The image-side surface **522** of the second lens element **520** has one inflection point.

(208) The third lens element **530** with negative refractive power has an object-side surface **531** being convex in a paraxial region thereof and an image-side surface **532** being concave in a paraxial region thereof. The third lens element **530** is made of plastic material and has the object-side surface **531** and the image-side surface **532** being both aspheric. The object-side surface **531** of the third lens element **530** has one inflection point.

(209) The fourth lens element **540** with negative refractive power has an object-side surface **541** being convex in a paraxial region thereof and an image-side surface **542** being concave in a paraxial region thereof. The fourth lens element **540** is made of plastic material and has the object-side surface **541** and the image-side surface **542** being both aspheric. The image-side surface **542** of the fourth lens element **540** has two inflection points.

(210) The fifth lens element **550** with positive refractive power has an object-side surface **551** being convex in a paraxial region thereof and an image-side surface **552** being concave in a paraxial region thereof. The fifth lens element **550** is made of plastic material and has the object-side surface **551** and the image-side surface **552** being both aspheric. The object-side surface **551** of the fifth lens element **550** has one inflection point. The image-side surface **552** of the fifth lens element **550** has three inflection points.

(211) The sixth lens element **560** with positive refractive power has an object-side surface **561** being convex in a paraxial region thereof and an image-side surface **562** being convex in a paraxial region thereof. The sixth lens element **560** is made of plastic material and has the object-side surface **561** and the image-side surface **562** being both aspheric. The object-side surface **561** of the sixth lens element **560** has three inflection points. The image-side surface **562** of the sixth lens element **560** has two inflection points.

(212) The seventh lens element **570** with negative refractive power has an object-side surface **571** being convex in a paraxial region thereof and an image-side surface **572** being concave in a paraxial region thereof. The seventh lens element **570** is made of plastic material and has the object-side surface **571** and the image-side surface **572** being both aspheric. The object-side surface **571** of the seventh lens element **570** has three inflection points. The image-side surface **572** of the seventh lens element **570** has two inflection points. The image-side surface **572** of the seventh lens element **570** has at least one critical point in an off-axis region thereof.

(213) The eighth lens element **580** with negative refractive power has an object-side surface **581** being convex in a paraxial region thereof and an image-side surface **582** being concave in a paraxial region thereof. The eighth lens element **580** is made of plastic material and has the object-side surface **581** and the image-side surface **582** being both aspheric. The object-side surface **581** of the eighth lens element **580** has four inflection points. The image-side surface **582** of the eighth lens element **580** has three inflection points.

(214) The filter **590** is made of glass material and located between the eighth lens element **580** and

the image surface **595**, and will not affect the focal length of the imaging lens system. The image sensor **599** is disposed on or near the image surface **595** of the imaging lens system.

(215) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **580** is V_8 , and a refractive index of the eighth lens element **580** is N_8 .

(216) The focal length of the imaging lens system is f , a curvature radius of an object-side surface of one lens element of the imaging lens system is R_o , a curvature radius of an image-side surface of the lens element is R_i , and one (the sixth lens element **560**) of the eight lens elements (**510**, **520**, **530**, **540**, **550**, **560**, **570** and **580**) satisfies $|f/R_o|+|f/R_i|<0.50$. In detail, a curvature radius of the object-side surface **561** of the sixth lens element **560** is R_{11} , a curvature radius of the image-side surface **562** of the sixth lens element **560** is R_{12} , and $|f/R_{11}|+|f/R_{12}|=0.19$.

(217) The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

(218) TABLE-US-00012 TABLE 9 5th Embodiment $f = 5.52$ mm, $F_{no} = 1.80$, $HFOV = 22.6$ deg.

Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	0 Object	Plano	Infinity
1 Lens 1	2.325 (ASP)	0.816	Plastic	1.545	56.1	6.30	2	6.320 (ASP)	0.535
3 Ape. Stop	Plano	—	—	—	—	—	—	—	—
4 Lens 2	2.873 (ASP)	0.575	Plastic	1.545	56.1	6.54	5	13.813 (ASP)	0.030
6 Lens 3	4.660 (ASP)	0.160	Plastic	1.669	19.5	—10.22	7	2.733 (ASP)	0.186
8 Lens 4	5.323 (ASP)	0.359	Plastic	1.639	23.5	—8.99	9	2.689 (ASP)	0.609
10 Lens 5	7.094 (ASP)	0.208	Plastic	1.639	23.5	105.91	11	7.835 (ASP)	0.199
12 Lens 6	58.599 (ASP)	0.406	Plastic	1.672	18.8	43.69	13	—58.673 (ASP)	0.375
14 Lens 7	1.514 (ASP)	0.325	Plastic	1.544	55.9	—42.55	15	1.314 (ASP)	0.366
16 Lens 8	142.857 (ASP)	0.761	Plastic	1.688	18.7	—5.24	17	3.511 (ASP)	0.200
18 Filter	Plano	0.110	Glass	1.517	64.2	—	19	Plano	0.107
20 Image	Plano	—	—	—	—	—	—	—	—

Note: Reference wavelength is 587.6 nm (d-line).

(219) TABLE-US-00013 TABLE 10 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k=$

—3.9245E-01	—1.9478E+00	4.3070E-01	—7.5158E+01	—1.0444E+01	—8.5708E+00	A4=	
1.1932E-04	3.9517E-03	2.4226E-02	3.0236E-02	—1.2099E-01	—8.2592E-02	A6=	
8.0959E-04	—1.3133E-02	—2.2118E-02	—3.8967E-02	2.3049E-01	2.4203E-01	A8=	
—8.6902E-04	7.1284E-03	—1.0389E-03	6.4043E-02	—1.8703E-01	—2.1529E-01	A10=	
1.9060E-04	—1.2500E-03	9.3825E-03	—6.6288E-02	8.6067E-02	9.0817E-02	A12=	
6.8738E-05	4.7403E-05	—3.5364E-03	3.0402E-02	—3.4271E-02	7.0803E-04	A14=	
—1.3004E-05	—	—5.3697E-03	1.4229E-02	—6.9636E-03	A16=	—	
—	—	—	—	—	—	—3.0092E-03	
—	Surface # 8	9	10	11	12	13	$k=$
—	—9.5921E+00	2.0205E+00	7.5969E-01	—9.0000E+01	5.0000E+01	9.0000E+01	A4=
—	—1.3190E-02	—3.6618E-02	—1.7908E-01	—1.9711E-01	—1.0436E-01	—1.0630E-01	A6=
—	2.8986E-02	—2.7624E-02	3.0368E-01	4.3135E-01	2.0625E-01	1.1738E-01	A8=
—	—8.0123E-02	5.8477E-03	—6.4496E-01	—6.7619E-01	—1.4579E-01	—5.0056E-02	A10=
—	1.0982E-01	—1.5558E-02	7.1197E-01	6.5792E-01	5.7647E-02	5.5756E-03	A12=
—	—5.6748E-02	5.5312E-02	—4.5868E-01	—3.8918E-01	—1.2967E-02	4.4246E-03	A14=
—	1.1499E-02	—5.3705E-02	1.5422E-01	1.2723E-01	1.5208E-03	—1.7704E-03	A16=
—	1.6290E-02	—2.1553E-02	—1.7366E-02	—7.8767E-05	1.8200E-04	Surface # 14	15
—	—4.2671E+00	—1.6654E+00	1.8959E+01	—5.4326E+01	A4=	—1.8360E-01	—3.6624E-01
—	—3.4287E-01	—4.2527E-01	A6=	9.1247E-02	4.0388E-01	6.3805E-01	6.3437E-01
—	—9.0141E-02	—3.8209E-01	—6.3537E-01	—5.1373E-01	A10=	7.8925E-02	2.4788E-01
—	3.9994E-01	2.6524E-01	A12=	—3.6860E-02	—1.0099E-01	—1.6456E-01	—9.0825E-02
—	8.3300E-03	2.4104E-02	4.3847E-02	2.0541E-02	A16=	—7.2271E-04	—2.9506E-03
—	—7.2748E-03	—2.9480E-03	A18=	—	1.0127E-04	6.8359E-04	2.4309E-04
—	—	—	A20=	—	7.9184E-06	—2.7833E-05	—8.7658E-06

(220) In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with

corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

(221) Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

(222) TABLE-US-00014 5th Embodiment f [mm] 5.52 EPD/BL 7.35 F_{no} 1.80 f /EPD 1.80 HFOV [deg.] 22.6 $(Y_{11} - \text{ImgH})/\text{EPD}$ -0.20 V_8 18.7 $f \times \tan(\text{HFOV})/\text{EPD}$ 0.75 V_{min} 18.7 $(|Y_{11} - Y_{82}| + \text{BL})/f$ 0.16 ΣV_i 272.0 $(Y_{82} - Y_{51})/(Y_{11} - Y_{51})$ 1.82 N_{max} 1.688 $(|Y_{11} - Y_{82}| + |Y_{82} - \text{ImgH}|)/f$ 0.11 $(V_i/N_i)_{min}$ 11.08 f/ImgH 2.31 $\Sigma \text{CT}/\Sigma \text{AT}$ 1.72 Y_{c72}/f 0.22 R_1/CT_1 2.85 $|f/R_1| + |f/R_2|$ 3.25 BL/CT_1 0.51 $|f/R_3| + |f/R_4|$ 2.32 R_8/f 0.49 $|f/R_5| + |f/R_6|$ 3.20 $(R_{11} + R_{12})/ -0.00063$ $|f/R_7| + |f/R_8|$ 3.09 $(R_{11} - R_{12}) (R_{13} - R_{14})/$ 0.07 $|f/R_9| + |f/R_{10}|$ 1.48 $(R_{13} + R_{14}) f/f_1$ 0.88 $|f/R_{11}| + |f/R_{12}|$ 0.19 f/f_5 0.05 $|f/R_{13}| + |f/R_{14}|$ 7.85 f_5/f_7 -2.49 $|f/R_{15}| + |f/R_{16}|$ 1.61 f_8/f_5 -0.05 V_1/N_1 36.30 $f/f_5 + f/f_6$ 0.18 V_2/N_2 36.30 ImgH [mm] 2.39 V_3/N_3 11.66 TL [mm] 6.12 V_4/N_4 14.34 TL/f 1.11 V_5/N_5 14.34 TL/EPD 2.00 V_6/N_6 11.24 SL/TL 0.78 V_7/N_7 36.23 BL/TD 0.07 V_8/N_8 11.08 EPD/ImgH 1.28 —

6th Embodiment

(223) FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment. In FIG. 11, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 699. The imaging lens system includes, in order from an object side to an image side, a first lens element 610, a second lens element 620, an aperture stop 600, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, a seventh lens element 670, an eighth lens element 680, a filter 690 and an image surface 695. The imaging lens system includes eight lens elements (610, 620, 630, 640, 650, 660, 670 and 680) with no additional lens element disposed between each of the adjacent eight lens elements.

(224) The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric. The object-side surface 611 of the first lens element 610 has one inflection point.

(225) The second lens element 620 with positive refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being convex in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric. The image-side surface 622 of the second lens element 620 has two inflection points.

(226) The third lens element 630 with negative refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

(227) The fourth lens element 640 with negative refractive power has an object-side surface 641 being convex in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric. The image-side surface 642 of the fourth lens element 640 has one inflection point.

(228) The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being concave in a paraxial region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric. The object-side surface 651 of the fifth lens element 650 has one inflection point. The image-side surface 652 of the fifth lens

element **650** has three inflection points.

(229) The sixth lens element **660** with positive refractive power has an object-side surface **661** being convex in a paraxial region thereof and an image-side surface **662** being convex in a paraxial region thereof. The sixth lens element **660** is made of plastic material and has the object-side surface **661** and the image-side surface **662** being both aspheric. The object-side surface **661** of the sixth lens element **660** has three inflection points. The image-side surface **662** of the sixth lens element **660** has two inflection points.

(230) The seventh lens element **670** with negative refractive power has an object-side surface **671** being convex in a paraxial region thereof and an image-side surface **672** being concave in a paraxial region thereof. The seventh lens element **670** is made of plastic material and has the object-side surface **671** and the image-side surface **672** being both aspheric. The object-side surface **671** of the seventh lens element **670** has three inflection points. The image-side surface **672** of the seventh lens element **670** has one inflection point. The image-side surface **672** of the seventh lens element **670** has at least one critical point in an off-axis region thereof.

(231) The eighth lens element **680** with positive refractive power has an object-side surface **681** being concave in a paraxial region thereof and an image-side surface **682** being convex in a paraxial region thereof. The eighth lens element **680** is made of plastic material and has the object-side surface **681** and the image-side surface **682** being both aspheric. The object-side surface **681** of the eighth lens element **680** has four inflection points. The image-side surface **682** of the eighth lens element **680** has six inflection points.

(232) The filter **690** is made of glass material and located between the eighth lens element **680** and the image surface **695**, and will not affect the focal length of the imaging lens system. The image sensor **699** is disposed on or near the image surface **695** of the imaging lens system.

(233) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to $V8/N8$, wherein an abbe number of the eighth lens element **680** is $V8$, and a refractive index of the eighth lens element **680** is $N8$.

(234) The focal length of the imaging lens system is f , a curvature radius of an object-side surface of one lens element of the imaging lens system is R_o , a curvature radius of an image-side surface of the lens element is R_i , and two (the sixth lens element **660** and the eighth lens element **680**) of the eight lens elements (**610**, **620**, **630**, **640**, **650**, **660**, **670** and **680**) satisfy $|f/R_o| + |f/R_i| < 0.50$. In detail, a curvature radius of the object-side surface **661** of the sixth lens element **660** is R_{11} , a curvature radius of the image-side surface **662** of the sixth lens element **660** is R_{12} , and $|f/R_{11}| + |f/R_{12}| = 0.14$; a curvature radius of the object-side surface **681** of the eighth lens element **680** is R_{15} , a curvature radius of the image-side surface **682** of the eighth lens element **680** is R_{16} , and $|f/R_{15}| + |f/R_{16}| = 0.31$.

(235) The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

(236) TABLE-US-00015 TABLE 11 6th Embodiment $f = 5.29$ mm, $Fno = 1.44$, $HFOV = 23.4$ deg.

Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	0	Object	Plano	Infinity
1	1.875 (ASP)	0.980	Plastic	1.545	56.1	6.13	2	3.487 (ASP)	0.030	3
Lens 2	3.167 (ASP)	0.758	Plastic	1.545	56.1	5.27	4	-28.233 (ASP)	0.056	5
Ape. Stop	Plano	0.016	6	Lens 3	8.835 (ASP)	0.160	Plastic	1.669	19.5	-6.98
7	3.032 (ASP)	0.247	8	Lens 4	6.128 (ASP)	0.200	Plastic	1.639	23.5	-9.65
9	3.033 (ASP)	0.440	10	Lens 5	4.670 (ASP)	0.174	Plastic	1.650	21.5	18.53
11	7.514 (ASP)	0.150	12	Lens 6	86.072 (ASP)	0.283	Plastic	1.672	18.8	54.59
13	-63.851 (ASP)	0.653	14	Lens 7	1.983 (ASP)	0.250	Plastic	1.544	55.9	-5.13
15	1.108 (ASP)	0.228	16	Lens 8	-77.715 (ASP)	0.441	Plastic	1.688	18.7	44.02
17	-21.842 (ASP)	0.200	18	Filter	Plano	0.110	Glass	1.517	64.2	—
19	Plano	0.170	20	Image	Plano	—	Note: Reference wavelength is 587.6 nm (d-line).			

(237) TABLE-US-00016 TABLE 12 Aspheric Coefficients Surface # 1 2 3 4 6 7 $k = -2.9987E-01$
 $-2.8667E-01$ $1.6017E-01$ $8.9940E+01$ $-1.6056E+01$ $-7.6812E+00$ $A4 = -9.0393E-03$

-3.5745E-03 -2.7682E-03 -1.1569E-02 -1.1218E-01 -6.9545E-02 A6= 1.4943E-02
 6.7680E-03 5.3631E-03 3.2951E-02 2.2318E-01 3.0679E-01 A8= -9.5400E-03
 -3.1125E-03 -2.3768E-03 -2.5115E-02 -2.3558E-01 -4.3066E-01 A10= 3.1104E-03
 4.0788E-04 2.4100E-04 9.5283E-03 1.3957E-01 3.5646E-01 A12= -3.9413E-04 — —
 -1.8658E-03 -4.1536E-02 -1.7387E-01 A14= — — — 1.4868E-04 4.9452E-03
 4.4190E-02 Surface # 8 9 10 11 12 13 k= -1.3391E+01 2.0407E+00 -1.1843E+01
 -4.2685E+01 -5.0000E+01 7.8245E+01 A4= -4.9793E-02 -7.0119E-02 -7.5137E-02
 -1.4123E-01 -1.0310E-01 -9.5361E-02 A6= 2.1320E-01 7.4146E-02 -8.9966E-02
 3.2300E-01 1.6570E-01 1.1258E-01 A8= -3.2457E-01 -1.3825E-02 1.2466E-01
 -6.2816E-01 -7.2489E-02 -6.3377E-02 A10= 2.0937E-01 -2.3542E-01 -2.1807E-01
 6.0871E-01 6.3589E-03 2.3809E-02 A12= -5.4262E-02 3.2231E-01 4.1894E-02
 -2.9492E-01 4.0891E-03 -4.6356E-03 A14= 5.9311E-03 -1.6462E-01 1.3178E-01
 6.9447E-02 -9.6065E-04 2.9119E-04 A16= — 2.9398E-02 -6.8777E-02 -6.3802E-03 — —
 Surface # 14 15 16 17 k= -1.0796E+01 -4.6820E+00 5.8009E+01 8.3614E+01 A4=
 -3.1062E-01 -2.7419E-01 -3.3896E-01 -4.5239E-01 A6= -1.9264E-02 1.2752E-01
 5.2119E-01 5.5455E-01 A8= 1.7837E-01 -2.2808E-02 -3.7993E-01 -3.0923E-01 A10=
 -1.0037E-01 -3.2557E-03 1.5597E-01 8.6629E-02 A12= 1.6696E-02 1.6441E-03
 -3.7626E-02 -9.6990E-03 A14= 1.7146E-03 -2.1007E-04 5.1899E-03 -6.5805E-04
 A16= -5.4247E-04 1.0790E-05 -3.6618E-04 2.5956E-04 A18= — — 9.5770E-06
 -1.7478E-05

(238) In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

(239) Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

(240) TABLE-US-00017 6th Embodiment f [mm] 5.29 EPD/BL 7.64 Fno 1.44 f/EPD 1.44 HFOV [deg.] 23.4 (Y11 - ImgH)/EPD -0.11 V8 18.7 f × tan(HFOV)/EPD 0.62 Vmin 18.7 (|Y11 - Y82| + BL)/f 0.15 ΣVi 270.0 (Y82 - Y51)/(Y11 - Y51) 1.40 Nmax 1.688 (|Y11 - Y82| + |Y82 - ImgH|)/f 0.08 (Vi/Ni)min 11.08 f/ImgH 2.22 ΣCT/ΣAT 1.78 Yc72/f 0.17 R1/CT1 1.91 f/R1| + |f/R2| 4.33 BL/CT1 0.49 |f/R3| + |f/R4| 1.86 R8/f 0.57 |f/R5| + |f/R6| 2.34 (R11 + R12)/ 0.15 |f/R7| + |f/R8| 2.61 (R11 - R12) (R13 - R14)/ 0.28 |f/R9| + |f/R10| 1.84 (R13 + R14) f/f1 0.86 |f/R11| + |f/R12| 0.14 f/f5 0.29 |f/R13| + |f/R14| 7.44 f5/f7 -3.61 |f/R15| + |f/R16| 0.31 f8/f5 2.38 V1/N1 36.30 f/f5 + f/f6 0.38 V2/N2 36.30 ImgH [mm] 2.38 V3/N3 11.66 TL [mm] 5.55 V4/N4 14.34 TL/f 1.05 V5/N5 13.01 TL/EPD 1.51 V6/N6 11.24 SL/TL 0.67 V7/N7 36.23 BL/TD 0.09 V8/N8 11.08 EPD/ImgH 1.54 — —

7th Embodiment

(241) FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment. In FIG. 13, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 799. The imaging lens system includes, in order from an object side to an image side, a first lens element 710, a second lens element 720, an aperture stop 700, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, a seventh lens element 770, an eighth lens element 780, a filter 790 and an image surface 795. The imaging lens system includes eight lens elements (710, 720, 730, 740, 750, 760, 770 and 780) with no additional lens element disposed between each of the adjacent eight lens elements.

(242) The first lens element 710 with positive refractive power has an object-side surface 711 being

convex in a paraxial region thereof and an image-side surface **712** being concave in a paraxial region thereof. The first lens element **710** is made of plastic material and has the object-side surface **711** and the image-side surface **712** being both aspheric. The object-side surface **711** of the first lens element **710** has one inflection point. The image-side surface **712** of the first lens element **710** has one inflection point.

(243) The second lens element **720** with positive refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being convex in a paraxial region thereof. The second lens element **720** is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric. The object-side surface **721** of the second lens element **720** has one inflection point. The image-side surface **722** of the second lens element **720** has two inflection points.

(244) The third lens element **730** with negative refractive power has an object-side surface **731** being concave in a paraxial region thereof and an image-side surface **732** being concave in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric. The object-side surface **731** of the third lens element **730** has one inflection point.

(245) The fourth lens element **740** with positive refractive power has an object-side surface **741** being convex in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric. The object-side surface **741** of the fourth lens element **740** has three inflection points. The image-side surface **742** of the fourth lens element **740** has one inflection point.

(246) The fifth lens element **750** with negative refractive power has an object-side surface **751** being concave in a paraxial region thereof and an image-side surface **752** being concave in a paraxial region thereof. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric. The image-side surface **752** of the fifth lens element **750** has two inflection points.

(247) The sixth lens element **760** with positive refractive power has an object-side surface **761** being concave in a paraxial region thereof and an image-side surface **762** being convex in a paraxial region thereof. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being both aspheric. The object-side surface **761** of the sixth lens element **760** has two inflection points. The image-side surface **762** of the sixth lens element **760** has two inflection points.

(248) The seventh lens element **770** with negative refractive power has an object-side surface **771** being convex in a paraxial region thereof and an image-side surface **772** being concave in a paraxial region thereof. The seventh lens element **770** is made of plastic material and has the object-side surface **771** and the image-side surface **772** being both aspheric. The object-side surface **771** of the seventh lens element **770** has two inflection points. The image-side surface **772** of the seventh lens element **770** has one inflection point. The image-side surface **772** of the seventh lens element **770** has at least one critical point in an off-axis region thereof.

(249) The eighth lens element **780** with negative refractive power has an object-side surface **781** being convex in a paraxial region thereof and an image-side surface **782** being concave in a paraxial region thereof. The eighth lens element **780** is made of plastic material and has the object-side surface **781** and the image-side surface **782** being both aspheric. The object-side surface **781** of the eighth lens element **780** has six inflection points. The image-side surface **782** of the eighth lens element **780** has four inflection points.

(250) The filter **790** is made of glass material and located between the eighth lens element **780** and the image surface **795**, and will not affect the focal length of the imaging lens system. The image sensor **799** is disposed on or near the image surface **795** of the imaging lens system.

(251) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-

th lens element is Ni, a minimum value of Vi/Ni is (Vi/Ni)min, and (Vi/Ni)min is equal to V8/N8, wherein an abbe number of the eighth lens element **780** is V8, and a refractive index of the eighth lens element **780** is N8.

(252) The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

(253) TABLE-US-00018 TABLE 13 7th Embodiment $f = 5.32$ mm, $Fno = 1.55$, $HFOV = 23.3$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Lens 1 1.911 (ASP) 0.980 Plastic 1.545 56.1 5.43 2 4.418 (ASP) 0.030 3 Lens 2 3.392 (ASP)
0.683 Plastic 1.545 56.1 5.23 4 -16.475 (ASP) 0.114 5 Ape. Stop Plano 0.043 6 Lens 3 -46.816
(ASP) 0.160 Plastic 1.669 19.5 -3.16 7 2.214 (ASP) 0.199 8 Lens 4 3.785 (ASP) 0.461 Plastic
1.639 23.5 6.09 9 132.551 (ASP) 0.435 10 Lens 5 -21.491 (ASP) 0.160 Plastic 1.650 21.5 -9.51
11 8.712 (ASP) 0.142 12 Lens 6 -44.889 (ASP) 0.346 Plastic 1.672 18.8 28.09 13 -13.328 (ASP)
0.481 14 Lens 7 3.748 (ASP) 0.195 Plastic 1.544 55.9 -4.83 15 1.515 (ASP) 0.180 16 Lens 8 2.939
(ASP) 0.434 Plastic 1.688 18.7 -11.29 17 2.004 (ASP) 0.200 18 Filter Plano 0.110 Glass 1.517
64.2 — 19 Plano 0.188 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(254) TABLE-US-00019 TABLE 14 Aspheric Coefficients Surface # 1 2 3 4 6 7 $k =$
-2.9619E-01 4.5805E-01 -1.8563E-01 -9.0000E+01 -9.0000E+01 -1.0548E+01 $A4 =$
-3.4463E-03 3.2884E-03 -9.1374E-03 2.8920E-02 -1.6164E-02 5.1733E-02 $A6 =$
1.0872E-02 -3.1477E-03 3.4998E-03 -1.3163E-02 8.6334E-03 -9.7889E-02 $A8 =$
-7.6822E-03 2.3344E-03 8.7197E-04 -4.0372E-03 -4.6348E-02 1.6607E-02 $A10 =$
2.7650E-03 -5.0328E-04 -4.2578E-04 4.9520E-03 6.9534E-02 4.6548E-02 $A12 =$
-4.0325E-04 — — -1.4605E-03 -3.5103E-02 2.5890E-03 $A14 =$ — — — 1.4177E-04
6.0059E-03 -7.8640E-03 Surface # 8 9 10 11 12 13 $k =$ -1.7482E+01 9.0000E+01
3.6643E+01 1.7836E+01 -5.0000E+01 5.3745E+01 $A4 =$ 1.8160E-02 -1.8922E-02
-7.1981E-02 -2.8384E-02 -7.3197E-02 -1.1356E-01 $A6 =$ 3.0824E-02 6.7119E-02
-2.7462E-01 -1.3255E-01 1.2676E-01 1.2830E-01 $A8 =$ -2.3577E-01 -2.4892E-01
3.0838E-01 1.6948E-01 -6.6003E-02 -9.5037E-02 $A10 =$ 2.6225E-01 2.8018E-01
-2.6693E-01 -9.9344E-02 1.5285E-02 5.5286E-02 $A12 =$ -8.9463E-02 -1.3636E-01
8.2093E-02 2.7386E-02 1.0886E-05 -1.6691E-02 $A14 =$ 4.3740E-03 1.8151E-02
4.1255E-02 2.7742E-03 -5.2513E-04 1.8728E-03 $A16 =$ — 2.4829E-03 -2.0828E-02
-2.2156E-03 — — Surface # 14 15 16 17 $k =$ -6.0047E+00 -6.8378E+00 -9.0000E+01
-9.0000E+01 $A4 =$ -3.9122E-01 -3.8687E-01 -3.8899E-01 -3.0485E-01 $A6 =$ 3.0048E-01
4.9620E-01 6.4983E-01 2.8636E-01 $A8 =$ -1.9286E-01 -3.8201E-01 -5.2621E-01
-1.0189E-01 $A10 =$ 1.0870E-01 1.7122E-01 2.4087E-01 -6.4814E-03 $A12 =$ -4.7081E-02
-4.3978E-02 -6.4934E-02 1.6352E-02 $A14 =$ 1.1921E-02 5.9117E-03 1.0130E-02
-5.1361E-03 $A16 =$ -1.2166E-03 -3.1991E-04 -8.3739E-04 6.9069E-04 $A18 =$ — —
2.8008E-05 -3.5209E-05

(255) In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

(256) Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

(257) TABLE-US-00020 7th Embodiment f [mm] 5.32 EPD/BL 6.90 Fno 1.55 f /EPD 1.55 $HFOV$ [deg.] 23.3 $(Y11 - ImgH)/EPD$ -0.12 V8 18.7 $f \times \tan(HFOV)/EPD$ 0.67 $Vmin$ 18.7 $(|Y11 - Y82| + BL)/f$ 0.14 ΣVi 270.0 $(Y82 - Y51)/(Y11 - Y51)$ 1.30 $Nmax$ 1.688 $(|Y11 - Y82| + |Y82 - ImgH|)/f$ 0.08 $(Vi/Ni)min$ 11.08 $f/ImgH$ 2.23 $\Sigma CT/\Sigma AT$ 2.11 $Yc72/f$ 0.18 $R1/CT1$ 1.95 $|f/R1| + |f/R2|$ 3.99 $BL/CT1$ 0.51 $|f/R3| + |f/R4|$ 1.89 $R8/f$ 24.90 $|f/R5| + |f/R6|$ 2.52 $(R11 + R12)/$ 1.84 $|f/R7| + |f/R8|$ 1.45 $(R11 - R12) (R13 - R14)/$ 0.42 $|f/R9| + |f/R10|$ 0.86 $(R13 + R14) f/f1$ 0.98 $|f/R11| + |f/R12|$

0.52 f/f5 -0.56 |f/R13| + |f/R14| 4.93 f5/f7 1.97 |f/R15| + |f/R16| 4.47 f8/f5 1.19 V1/N1 36.30 f/f5 + f/f6 -0.37 V2/N2 36.30 ImgH [mm] 2.39 V3/N3 11.66 TL [mm] 5.54 V4/N4 14.34 TL/f 1.04 V5/N5 13.01 TL/EPD 1.61 V6/N6 11.24 SL/TL 0.67 V7/N7 36.23 BL/TD 0.10 V8/N8 11.08 EPD/ImgH 1.44 — —

8th Embodiment

(258) FIG. 15 is a schematic view of an image capturing unit according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment. In FIG. 15, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 899. The imaging lens system includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, a seventh lens element 870, an eighth lens element 880, a filter 890 and an image surface 895. The imaging lens system includes eight lens elements (810, 820, 830, 840, 850, 860, 870 and 880) with no additional lens element disposed between each of the adjacent eight lens elements.

(259) The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being convex in a paraxial region thereof. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric. The image-side surface 812 of the first lens element 810 has one inflection point.

(260) The second lens element 820 with negative refractive power has an object-side surface 821 being convex in a paraxial region thereof and an image-side surface 822 being concave in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric. The object-side surface 821 of the second lens element 820 has three inflection points. The image-side surface 822 of the second lens element 820 has one inflection point.

(261) The third lens element 830 with positive refractive power has an object-side surface 831 being convex in a paraxial region thereof and an image-side surface 832 being concave in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric. The object-side surface 831 of the third lens element 830 has two inflection points.

(262) The fourth lens element 840 with positive refractive power has an object-side surface 841 being convex in a paraxial region thereof and an image-side surface 842 being concave in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being both aspheric. The object-side surface 841 of the fourth lens element 840 has one inflection point. The image-side surface 842 of the fourth lens element 840 has one inflection point.

(263) The fifth lens element 850 with positive refractive power has an object-side surface 851 being convex in a paraxial region thereof and an image-side surface 852 being convex in a paraxial region thereof. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being both aspheric. The object-side surface 851 of the fifth lens element 850 has one inflection point. The image-side surface 852 of the fifth lens element 850 has two inflection points.

(264) The sixth lens element 860 with negative refractive power has an object-side surface 861 being concave in a paraxial region thereof and an image-side surface 862 being concave in a paraxial region thereof. The sixth lens element 860 is made of plastic material and has the object-side surface 861 and the image-side surface 862 being both aspheric. The object-side surface 861 of the sixth lens element 860 has two inflection points. The image-side surface 862 of the sixth lens element 860 has one inflection point.

(265) The seventh lens element **870** with negative refractive power has an object-side surface **871** being convex in a paraxial region thereof and an image-side surface **872** being concave in a paraxial region thereof. The seventh lens element **870** is made of plastic material and has the object-side surface **871** and the image-side surface **872** being both aspheric. The object-side surface **871** of the seventh lens element **870** has one inflection point. The image-side surface **872** of the seventh lens element **870** has one inflection point. The image-side surface **872** of the seventh lens element **870** has at least one critical point in an off-axis region thereof.

(266) The eighth lens element **880** with positive refractive power has an object-side surface **881** being convex in a paraxial region thereof and an image-side surface **882** being concave in a paraxial region thereof. The eighth lens element **880** is made of plastic material and has the object-side surface **881** and the image-side surface **882** being both aspheric. The object-side surface **881** of the eighth lens element **880** has two inflection points.

(267) The filter **890** is made of glass material and located between the eighth lens element **880** and the image surface **895**, and will not affect the focal length of the imaging lens system. The image sensor **899** is disposed on or near the image surface **895** of the imaging lens system.

(268) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **880** is V_8 , and a refractive index of the eighth lens element **880** is N_8 .

(269) The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

(270) TABLE-US-00021 TABLE 15 8th Embodiment $f = 4.87$ mm, $F_{no} = 1.36$, $HFOV = 21.8$ deg.
 Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
 1 Ape. Stop Plano -1.028 2 Lens 1 1.934 (ASP) 1.236 Plastic 1.545 56.0 3.18 3 -13.003 (ASP)
 0.030 4 Lens 2 43.718 (ASP) 0.170 Plastic 1.614 26.0 -3.23 5 1.894 (ASP) 0.226 6 Lens 3 1.962
 (ASP) 0.270 Plastic 1.582 30.2 8.87 7 3.003 (ASP) 0.050 8 Lens 4 3.234 (ASP) 0.543 Plastic 1.530
 55.8 21.41 9 4.260 (ASP) 0.698 10 Lens 5 12.479 (ASP) 0.180 Plastic 1.584 28.2 7.56 11 -6.807
 (ASP) 0.035 12 Lens 6 -21.586 (ASP) 0.394 Plastic 1.614 26.0 -7.60 13 5.992 (ASP) 0.198 14
 Lens 7 5.941 (ASP) 0.601 Plastic 1.582 30.2 -6.79 15 2.286 (ASP) 0.095 16 Lens 8 12.070 (ASP)
 0.209 Plastic 1.669 19.5 30.78 17 28.978 (ASP) 0.300 18 Filter Plano 0.145 Glass 1.517 64.2 — 19
 Plano 0.258 20 Image Plano — Note: Reference wavelength is 587.6 nm (d-line).

(271) TABLE-US-00022 TABLE 16 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k =$
 $-4.0887E-01$ $2.4806E+01$ $-9.0000E+01$ $-8.1291E+00$ $-1.5828E+01$ $-2.9794E+01$ $A_4 =$
 $2.1793E-03$ $-7.4888E-03$ $-9.5426E-02$ $2.9820E-02$ $2.1306E-01$ $1.4667E-01$ $A_6 =$
 $4.0592E-03$ $3.9245E-02$ $1.5810E-01$ $-3.6952E-02$ $-4.5215E-01$ $-3.2185E-01$ $A_8 =$
 $-3.4405E-03$ $-2.8167E-02$ $-1.1192E-01$ $1.1117E-01$ $4.6739E-01$ $3.6335E-01$ $A_{10} =$
 $1.8385E-03$ $9.6744E-03$ $4.3312E-02$ $-9.0744E-02$ $-2.7920E-01$ $-2.3181E-01$ $A_{12} =$
 $-4.5180E-04$ $-1.6484E-03$ $-9.0000E-03$ $2.5488E-02$ $8.9907E-02$ $7.1271E-02$ $A_{14} =$
 $3.9578E-05$ $1.1183E-04$ $8.3006E-04$ $3.6688E-03$ $-8.7261E-03$ $-8.0636E-04$ $A_{16} =$ — —
 $-1.3960E-05$ $-2.1202E-03$ $-1.0200E-03$ $-2.5882E-03$ Surface # 8 9 10 11 12 13 $k =$
 $-5.6238E-01$ $-1.7584E+00$ $7.7610E+01$ $2.1137E+01$ $4.0565E+01$ $-8.2800E+01$ $A_4 =$
 $-1.7222E-04$ $-3.6150E-02$ $-8.9824E-02$ $2.3488E-01$ $2.2626E-01$ $-1.5351E-01$ $A_6 =$
 $-3.9382E-02$ $3.6464E-03$ $1.6830E-02$ $-8.2089E-01$ $-8.3108E-01$ $9.1000E-02$ $A_8 =$
 $1.3858E-01$ $1.3204E-02$ $-3.4820E-01$ $9.5210E-01$ $1.1720E+00$ $-1.0561E-01$ $A_{10} =$
 $-1.5548E-01$ $-3.9764E-02$ $5.3314E-01$ $-4.7173E-01$ $-8.6223E-01$ $5.0282E-02$ $A_{12} =$
 $7.5781E-02$ $2.3860E-02$ $-3.2357E-01$ $1.0104E-01$ $3.3253E-01$ $-8.3698E-03$ $A_{14} =$
 $-1.3615E-02$ $-4.6269E-03$ $6.9518E-02$ $-7.5991E-03$ $-5.4134E-02$ $2.2299E-04$ Surface # 14
 15 16 17 $k =$ $1.3977E+01$ $-3.8327E-01$ $3.3367E+01$ $-8.2360E+01$ $A_4 =$ $-3.8915E-01$
 $-2.5620E-01$ $-2.8843E-02$ $-2.6637E-02$ $A_6 =$ $2.1273E-01$ $1.7862E-01$ $6.0179E-02$
 $5.2147E-02$ $A_8 =$ $-3.1668E-02$ $-1.2081E-01$ $-8.3670E-02$ $-4.9687E-02$ $A_{10} =$ $-9.8250E-02$

5.2131E-02 5.2508E-02 2.5677E-02 A12= 8.3916E-02 -1.2452E-02 -1.6380E-02
-6.8438E-03 A14= -2.5052E-02 1.4066E-03 2.5226E-03 8.8750E-04 A16= 2.3496E-03
-4.9718E-05 -1.5397E-04 -4.3471E-05

(272) In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

(273) Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

(274) TABLE-US-00023 8th Embodiment f [mm] 4.87 EPD/BL 5.09 Fno 1.36 f/EPD 1.36 HFOV [deg.] 21.8 (Y11 - ImgH)/EPD -0.06 V8 19.5 f × tan(HFOV)/EPD 0.54 Vmin 19.5 (|Y11 - Y82| + BL)/f 0.15 ΣVi 271.9 (Y82 - Y51)/(Y11 - Y51) 1.01 Nmax 1.669 (|Y11 - Y82| + |Y82 - ImgH|)/f 0.04 (Vi/Ni)min 11.66 f/ImgH 2.43 ΣCT/ΣAT 2.70 Yc72/f 0.18 R1/CT1 1.56 |f/R1| + |f/R2| 2.89 BL/CT1 0.57 |f/R3| + |f/R4| 2.68 R8/f 0.88 |f/R5| + |f/R6| 4.10 (R11 + R12)/ 0.57 |f/R7| + |f/R8| 2.65 (R11 - R12) (R13 - R14)/ 0.44 |f/R9| + |f/R10| 1.10 (R13 + R14) f/f1 1.53 |f/R11| + |f/R12| 1.04 f/f5 0.64 |f/R13| + |f/R14| 2.95 f5/f7 -1.11 |f/R15| + |f/R16| 0.57 f8/f5 4.07 V1/N1 36.27 f/f5 + f/f6 0.0030 V2/N2 16.09 ImgH [mm] 2.00 V3/N3 19.11 TL [mm] 5.64 V4/N4 36.45 TL/f 1.16 V5/N5 17.80 TL/EPD 1.58 V6/N6 16.09 SL/TL 0.82 V7/N7 19.11 BL/TD 0.14 V8/N8 11.66 EPD/ImgH 1.79 — —

9th Embodiment

(275) FIG. 17 is a schematic view of an image capturing unit according to the 9th embodiment of the present disclosure. FIG. 18 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment. In FIG. 17, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 999. The imaging lens system includes, in order from an object side to an image side, an aperture stop 900, a first lens element 910, a second lens element 920, a third lens element 930, a stop 901, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, a seventh lens element 970, an eighth lens element 980, a filter 990 and an image surface 995. The imaging lens system includes eight lens elements (910, 920, 930, 940, 950, 960, 970 and 980) with no additional lens element disposed between each of the adjacent eight lens elements.

(276) The first lens element 910 with positive refractive power has an object-side surface 911 being convex in a paraxial region thereof and an image-side surface 912 being convex in a paraxial region thereof. The first lens element 910 is made of plastic material and has the object-side surface 911 and the image-side surface 912 being both aspheric. The image-side surface 912 of the first lens element 910 has one inflection point.

(277) The second lens element 920 with negative refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being concave in a paraxial region thereof. The second lens element 920 is made of plastic material and has the object-side surface 921 and the image-side surface 922 being both aspheric. The object-side surface 921 of the second lens element 920 has three inflection points.

(278) The third lens element 930 with positive refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being concave in a paraxial region thereof. The third lens element 930 is made of plastic material and has the object-side surface 931 and the image-side surface 932 being both aspheric.

(279) The fourth lens element 940 with negative refractive power has an object-side surface 941 being convex in a paraxial region thereof and an image-side surface 942 being concave in a paraxial region thereof. The fourth lens element 940 is made of plastic material and has the object-side surface 941 and the image-side surface 942 being both aspheric. The object-side surface 941 of

the fourth lens element **940** has one inflection point. The image-side surface **942** of the fourth lens element **940** has one inflection point.

(280) The fifth lens element **950** with positive refractive power has an object-side surface **951** being convex in a paraxial region thereof and an image-side surface **952** being convex in a paraxial region thereof. The fifth lens element **950** is made of plastic material and has the object-side surface **951** and the image-side surface **952** being both aspheric. The object-side surface **951** of the fifth lens element **950** has three inflection points. The image-side surface **952** of the fifth lens element **950** has one inflection point.

(281) The sixth lens element **960** with negative refractive power has an object-side surface **961** being convex in a paraxial region thereof and an image-side surface **962** being concave in a paraxial region thereof. The sixth lens element **960** is made of plastic material and has the object-side surface **961** and the image-side surface **962** being both aspheric. The object-side surface **961** of the sixth lens element **960** has one inflection point. The image-side surface **962** of the sixth lens element **960** has one inflection point.

(282) The seventh lens element **970** with negative refractive power has an object-side surface **971** being convex in a paraxial region thereof and an image-side surface **972** being concave in a paraxial region thereof. The seventh lens element **970** is made of plastic material and has the object-side surface **971** and the image-side surface **972** being both aspheric. The object-side surface **971** of the seventh lens element **970** has two inflection points. The image-side surface **972** of the seventh lens element **970** has one inflection point. The image-side surface **972** of the seventh lens element **970** has at least one critical point in an off-axis region thereof.

(283) The eighth lens element **980** with positive refractive power has an object-side surface **981** being convex in a paraxial region thereof and an image-side surface **982** being convex in a paraxial region thereof. The eighth lens element **980** is made of plastic material and has the object-side surface **981** and the image-side surface **982** being both aspheric. The object-side surface **981** of the eighth lens element **980** has three inflection points. The image-side surface **982** of the eighth lens element **980** has two inflection points.

(284) The filter **990** is made of glass material and located between the eighth lens element **980** and the image surface **995**, and will not affect the focal length of the imaging lens system. The image sensor **999** is disposed on or near the image surface **995** of the imaging lens system.

(285) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_8/N_8 , wherein an abbe number of the eighth lens element **980** is V_8 , and a refractive index of the eighth lens element **980** is N_8 .

(286) The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

(287) TABLE-US-00024 TABLE 17 9th Embodiment $f = 6.15$ mm, $F_{no} = 1.59$, $HFOV = 18.8$ deg.
Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity
1 Ape. Stop Plano -1.173 2 Lens 1 1.935 (ASP) 1.470 Plastic 1.545 56.0 3.14 3 -10.786 (ASP)
 0.040 4 Lens 2 87.418 (ASP) 0.170 Plastic 1.614 26.0 -4.57 5 2.718 (ASP) 0.260 6 Lens 3 2.532
(ASP) 0.160 Plastic 1.582 30.2 356.57 7 2.504 (ASP) 0.450 8 Stop Plano -0.400 9 Lens 4 2.371
(ASP) 0.514 Plastic 1.530 55.8 -287.79 10 2.160 (ASP) 0.621 11 Lens 5 8.576 (ASP) 0.180 Plastic
 1.584 28.2 7.32 12 -8.459 (ASP) 0.049 13 Lens 6 45.591 (ASP) 0.170 Plastic 1.614 26.0 -5.99 14
 3.395 (ASP) 0.568 15 Lens 7 5.412 (ASP) 0.175 Plastic 1.582 30.2 -5.23 16 1.927 (ASP) 0.168 17
Lens 8 12.327 (ASP) 0.606 Plastic 1.669 19.5 10.88 18 -17.395 (ASP) 0.300 19 Filter Plano 0.145
Glass 1.517 64.2 — 20 Plano 0.253 21 Image Plano — Note: Reference wavelength is 587.6 nm
(d-line). An effective radius of the stop 901 (Surface 8) is 1.301 mm.

(288) TABLE-US-00025 TABLE 18 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k =$
 $-4.5860E-01$ $2.8791E+01$ $-8.9603E+01$ $-7.8897E+00$ $-1.9952E+01$ $-2.1118E+01$ $A_4 =$
 $2.3046E-03$ $-7.5505E-02$ $-1.0641E-01$ $-3.6631E-03$ $3.6026E-03$ $7.0356E-02$ $A_6 =$

5.5485E-03 1.0539E-01 1.3737E-01 1.1723E-01 7.9698E-02 -2.2493E-01 A8=
-5.3536E-03 -6.2893E-02 -7.7905E-02 -1.5940E-01 -1.5189E-01 5.1587E-01 A10=
3.0692E-03 2.1028E-02 2.5447E-02 1.3071E-01 1.2113E-01 -5.0920E-01 A12=
-8.6766E-04 -3.7284E-03 -3.9228E-03 -5.3401E-02 -4.7559E-02 2.3455E-01 A14=
8.6524E-05 2.7703E-04 1.7136E-04 9.5932E-03 8.8955E-03 -3.9828E-02 Surface # 9 10 11
12 13 14 k= 4.4541E-01 -2.3276E+00 -2.3063E+01 2.2304E+01 -9.0000E+01 -8.2800E+01
A4= 2.6219E-02 -2.4030E-02 -1.4750E-02 1.5878E-01 -9.8609E-02 -5.1954E-02 A6=
-2.9684E-01 -7.7358E-02 -3.7102E-01 -3.4130E-01 2.2516E-01 -1.9037E-02 A8=
6.9634E-01 1.1534E-01 3.2261E-01 1.5569E-01 -3.1129E-01 6.4470E-02 A10=
-6.7324E-01 -8.4656E-02 -3.5928E-01 1.6088E-02 2.4974E-01 -4.9426E-02 A12=
3.1218E-01 1.9508E-02 2.9734E-01 -3.8652E-03 -1.1711E-01 1.6523E-02 A14=
-5.6932E-02 — -8.3881E-02 — 2.2259E-02 -2.3234E-03 Surface # 15 16 17 18 k=
1.1890E+01 -1.0104E+00 3.4255E+01 -8.2360E+01 A4= -2.9706E-01 -2.9732E-01
-1.5346E-01 -2.0739E-01 A6= -1.3862E-01 1.1044E-01 2.7000E-01 1.6211E-01 A8=
3.6718E-01 2.5399E-02 -2.8846E-01 -8.8193E-02 A10= -3.1060E-01 -9.1938E-02
1.6347E-01 3.0802E-02 A12= 1.2481E-01 6.0875E-02 -4.9494E-02 -6.1821E-03 A14=
-1.9124E-02 -1.7009E-02 7.5349E-03 7.1398E-04 A16= 1.1144E-04 1.7502E-03
-4.0380E-04 -4.3935E-05 A18= — — -1.1557E-05 —

(289) In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

(290) Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

(291) TABLE-US-00026 9th Embodiment f [mm] 6.15 EPD/BL 5.54 Fno 1.59 f/EPD 1.59 HFOV [deg.] 18.8 (Y11 - ImgH)/EPD -0.05 V8 19.5 f × tan(HFOV)/EPD 0.54 Vmin 19.5 (|Y11 - Y82| + BL)/f 0.12 ΣVi 271.9 (Y82 - Y51)/(Y11 - Y51) 0.95 Nmax 1.669 (|Y11 - Y82| + |Y82 - ImgH|)/f 0.05 (Vi/Ni)min 11.66 f/ImgH 2.87 ΣCT/ΣAT 1.96 Yc72/f 0.13 R1/CT1 1.32 |f/R1| + |f/R2| 3.75 BL/CT1 0.48 |f/R3| + |f/R4| 2.33 R8/f 0.35 |f/R5| + |f/R6| 4.89 (R11 + R12)/ 1.16 |f/R7| + |f/R8| 5.44 (R11 - R12) (R13 - R14)/ 0.47 |f/R9| + |f/R10| 1.44 (R13 + R14) f/f1 1.96 |f/R11| + |f/R12| 1.95 f/f5 0.84 |f/R13| + |f/R14| 4.33 f5/f7 -1.40 |f/R15| + |f/R16| 0.85 f8/f5 1.49 V1/N1 36.27 f/f5 + f/f6 -0.19 V2/N2 16.09 ImgH [mm] 2.14 V3/N3 19.11 TL [mm] 5.90 V4/N4 36.45 TL/f 0.96 V5/N5 17.80 TL/EPD 1.52 V6/N6 16.09 SL/TL 0.80 V7/N7 19.11 BL/TD 0.13 V8/N8 11.66 EPD/ImgH 1.81 — —

10th Embodiment

(292) FIG. 19 is a schematic view of an image capturing unit according to the 10th embodiment of the present disclosure. FIG. 20 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 10th embodiment. In FIG. 19, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 1099. The imaging lens system includes, in order from an object side to an image side, a first lens element 1010, an aperture stop 1000, a second lens element 1020, a third lens element 1030, a stop 1001, a fourth lens element 1040, a fifth lens element 1050, a sixth lens element 1060, a seventh lens element 1070, an eighth lens element 1080, a filter 1090 and an image surface 1095. The imaging lens system includes eight lens elements (1010, 1020, 1030, 1040, 1050, 1060, 1070 and 1080) with no additional lens element disposed between each of the adjacent eight lens elements.

(293) The first lens element 1010 with positive refractive power has an object-side surface 1011 being convex in a paraxial region thereof and an image-side surface 1012 being convex in a paraxial region thereof. The first lens element 1010 is made of plastic material and has the object-

side surface **1011** and the image-side surface **1012** being both aspheric. The image-side surface **1012** of the first lens element **1010** has one inflection point.

(294) The second lens element **1020** with negative refractive power has an object-side surface **1021** being concave in a paraxial region thereof and an image-side surface **1022** being convex in a paraxial region thereof. The second lens element **1020** is made of plastic material and has the object-side surface **1021** and the image-side surface **1022** being both aspheric. The object-side surface **1021** of the second lens element **1020** has one inflection point. The image-side surface **1022** of the second lens element **1020** has one inflection point.

(295) The third lens element **1030** with negative refractive power has an object-side surface **1031** being concave in a paraxial region thereof and an image-side surface **1032** being concave in a paraxial region thereof. The third lens element **1030** is made of plastic material and has the object-side surface **1031** and the image-side surface **1032** being both aspheric. The object-side surface **1031** of the third lens element **1030** has three inflection points. The image-side surface **1032** of the third lens element **1030** has one inflection point.

(296) The fourth lens element **1040** with positive refractive power has an object-side surface **1041** being convex in a paraxial region thereof and an image-side surface **1042** being concave in a paraxial region thereof. The fourth lens element **1040** is made of plastic material and has the object-side surface **1041** and the image-side surface **1042** being both aspheric. The image-side surface **1042** of the fourth lens element **1040** has one inflection point.

(297) The fifth lens element **1050** with positive refractive power has an object-side surface **1051** being convex in a paraxial region thereof and an image-side surface **1052** being convex in a paraxial region thereof. The fifth lens element **1050** is made of plastic material and has the object-side surface **1051** and the image-side surface **1052** being both aspheric. The object-side surface **1051** of the fifth lens element **1050** has one inflection point. The image-side surface **1052** of the fifth lens element **1050** has one inflection point.

(298) The sixth lens element **1060** with negative refractive power has an object-side surface **1061** being concave in a paraxial region thereof and an image-side surface **1062** being concave in a paraxial region thereof. The sixth lens element **1060** is made of plastic material and has the object-side surface **1061** and the image-side surface **1062** being both aspheric. The object-side surface **1061** of the sixth lens element **1060** has one inflection point. The image-side surface **1062** of the sixth lens element **1060** has one inflection point.

(299) The seventh lens element **1070** with negative refractive power has an object-side surface **1071** being concave in a paraxial region thereof and an image-side surface **1072** being concave in a paraxial region thereof. The seventh lens element **1070** is made of plastic material and has the object-side surface **1071** and the image-side surface **1072** being both aspheric. The image-side surface **1072** of the seventh lens element **1070** has one inflection point. The image-side surface **1072** of the seventh lens element **1070** has at least one critical point in an off-axis region thereof.

(300) The eighth lens element **1080** with positive refractive power has an object-side surface **1081** being convex in a paraxial region thereof and an image-side surface **1082** being convex in a paraxial region thereof. The eighth lens element **1080** is made of plastic material and has the object-side surface **1081** and the image-side surface **1082** being both aspheric. The object-side surface **1081** of the eighth lens element **1080** has two inflection points. The image-side surface **1082** of the eighth lens element **1080** has one inflection point.

(301) The filter **1090** is made of glass material and located between the eighth lens element **1080** and the image surface **1095**, and will not affect the focal length of the imaging lens system. The image sensor **1099** is disposed on or near the image surface **1095** of the imaging lens system.

(302) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_2/N_2 , wherein an abbe number of the second lens element **1020** is V_2 , and a refractive index of the second lens element **1020** is N_2 .

(303) The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

(304) TABLE-US-00027 TABLE 19 10th Embodiment $f = 5.67$ mm, $Fno = 1.56$, $HFOV = 20.1$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 2.037 (ASP) 1.190 Plastic 1.545 56.0 3.21 2 -9.729 (ASP) -0.063 3 Ape. Stop Plano 0.093 4 Lens 2 -9.427 (ASP) 0.185 Plastic 1.660 20.4 -16.40 5 -73.720 (ASP) 0.166 6 Lens 3 -31.250 (ASP) 0.160 Plastic 1.582 30.2 -4.82 7 3.087 (ASP) 0.488 8 Stop Plano -0.420 9 Lens 4 2.163 (ASP) 0.601 Plastic 1.530 55.8 12.64 10 2.885 (ASP) 0.707 11 Lens 5 6.102 (ASP) 0.180 Plastic 1.584 28.2 5.20 12 -5.983 (ASP) 0.035 13 Lens 6 -19.306 (ASP) 0.233 Plastic 1.614 26.0 -5.81 14 4.392 (ASP) 0.718 15 Lens 7 -22.065 (ASP) 0.409 Plastic 1.534 55.9 -4.75 16 2.886 (ASP) 0.065 17 Lens 8 12.419 (ASP) 0.387 Plastic 1.650 21.5 12.39 18 -22.684 (ASP) 0.300 19 Filter Plano 0.145 Glass 1.517 64.2 — 20 Plano 0.259 21 Image Plano — Note: Reference wavelength is 587.6 nm (d-line). An effective radius of the stop 1001 (Surface 8) is 1.298 mm.

(305) TABLE-US-00028 TABLE 20 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$
-6.0388E-01 2.5859E+01 -8.8327E+01 9.0000E+01 -9.0000E+01 -3.8863E+00 $A4 =$
5.1124E-03 -3.9301E-03 -5.3078E-02 -8.9970E-02 3.6956E-02 2.1089E-01 $A6 =$
1.0492E-03 3.9686E-02 1.2654E-01 1.9925E-01 -6.7688E-02 -2.7893E-01 $A8 =$
-1.3832E-03 -3.7029E-02 -8.7424E-02 -1.5907E-01 6.3790E-02 2.1480E-01 $A10 =$
9.3876E-04 1.6678E-02 3.5731E-02 8.7062E-02 -2.6192E-02 -8.4078E-02 $A12 =$
-5.2700E-04 -3.7331E-03 -8.2279E-03 -2.8741E-02 5.0722E-03 2.1428E-02 $A14 =$
8.7629E-05 3.4792E-04 7.8552E-04 4.0276E-03 -3.0301E-04 -3.7881E-03 Surface # 9 10 11
12 13 14 $k = 4.0970E-01 -2.6123E+00 -4.5273E+01 1.2338E+01 5.0336E+01 -8.2800E+01$
 $A4 = 3.7133E-02 -7.8789E-02 -7.5384E-02 1.8170E-01 7.1945E-02 -7.2376E-02$ $A6 =$
-4.5057E-02 5.7975E-02 -1.8642E-01 -5.7860E-01 -3.5314E-01 -5.7692E-02 $A8 =$
2.0861E-02 -2.8864E-02 1.0991E-01 6.8105E-01 5.3027E-01 8.6130E-02 $A10 =$
6.4853E-03 -3.9088E-03 -1.7384E-01 -5.6524E-01 -4.6360E-01 -8.7622E-02 $A12 =$
-4.3362E-03 3.2084E-03 1.8136E-01 3.2872E-01 2.1671E-01 4.5144E-02 $A14 =$ — —
-5.3764E-02 -7.7438E-02 -3.9534E-02 -8.2883E-03 Surface # 15 16 17 18 $k =$
-9.0000E+01 3.1752E-01 3.4255E+01 -8.2360E+01 $A4 = -2.5367E-01 -2.3758E-01$
-5.0038E-02 -6.0346E-02 $A6 = 6.8531E-02 1.1153E-01 5.7099E-02 7.2670E-02$ $A8 =$
3.9419E-02 -3.5665E-02 -4.0208E-02 -4.7358E-02 $A10 = -6.8865E-02 -2.8165E-05$
1.3143E-02 1.6165E-02 $A12 = 3.7204E-02 4.1722E-03 -1.7348E-03 -3.1102E-03$ $A14 =$
-8.5730E-03 -1.2749E-03 3.0961E-05 3.8782E-04 $A16 = 6.8150E-04 1.2104E-04$
6.9776E-06 -2.6349E-05

(306) In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

(307) Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following conditions:

(308) TABLE-US-00029 10th Embodiment f [mm] 5.67 EPD/BL 5.16 Fno 1.56 f /EPD 1.56 $HFOV$ [deg.] 20.1 $(Y11 - ImgH)/EPD$ -0.08 $V8$ 21.5 $f \times \tan(HFOV)/EPD$ 0.57 $Vmin$ 20.4 $(|Y11 - Y82| + BL)/f$ 0.14 ΣVi 294.0 $(Y82 - Y51)/(Y11 - Y51)$ 1.16 $Nmax$ 1.660 $(|Y11 - Y82| + |Y82 - ImgH|)/f$ 0.05 $(Vi/Ni)min$ 12.29 $f/ImgH$ 2.65 $\Sigma CT/\Sigma AT$ 1.87 $Yc72/f$ 0.13 $R1/CT1$ 1.71 $|f/R1| + |f/R2|$ 3.37 $BL/CT1$ 0.59 $|f/R3| + |f/R4|$ 0.68 $R8/f$ 0.51 $|f/R5| + |f/R6|$ 2.02 $(R11 + R12)/0.63$ $|f/R7| + |f/R8|$ 4.59 $(R11 - R12)(R13 - R14)/1.30$ $|f/R9| + |f/R10|$ 1.88 $(R13 + R14)f/f1$ 1.77 $|f/R11| + |f/R12|$ 1.58 $f/f5$ 1.09 $|f/R13| + |f/R14|$ 2.22 $f5/f7$ -1.09 $|f/R15| + |f/R16|$ 0.71 $f8/f5$ 2.38 $V1/N1$ 36.27 $f/f5 + f/f6$ 0.11 $V2/N2$ 12.29 $ImgH$ [mm] 2.14 $V3/N3$ 19.11 TL [mm] 5.84 $V4/N4$ 36.45 TL/f 1.03 $V5/N5$ 17.80 TL/EPD 1.61 $V6/N6$ 16.09 SL/TL 0.81 $V7/N7$ 36.46 BL/TD 0.14 $V8/N8$ 13.01 $EPD/ImgH$

11th Embodiment

(309) FIG. **21** is a schematic view of an image capturing unit according to the 11th embodiment of the present disclosure. FIG. **22** shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 11th embodiment. In FIG. **21**, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor **1199**. The imaging lens system includes, in order from an object side to an image side, a first lens element **1110**, an aperture stop **1100**, a second lens element **1120**, a third lens element **1130**, a stop **1101**, a fourth lens element **1140**, a fifth lens element **1150**, a sixth lens element **1160**, a seventh lens element **1170**, an eighth lens element **1180**, a filter **1190** and an image surface **1195**. The imaging lens system includes eight lens elements (**1110**, **1120**, **1130**, **1140**, **1150**, **1160**, **1170** and **1180**) with no additional lens element disposed between each of the adjacent eight lens elements.

(310) The first lens element **1110** with positive refractive power has an object-side surface **1111** being convex in a paraxial region thereof and an image-side surface **1112** being convex in a paraxial region thereof. The first lens element **1110** is made of plastic material and has the object-side surface **1111** and the image-side surface **1112** being both aspheric. The object-side surface **1111** of the first lens element **1110** has two inflection points. The image-side surface **1112** of the first lens element **1110** has one inflection point.

(311) The second lens element **1120** with negative refractive power has an object-side surface **1121** being concave in a paraxial region thereof and an image-side surface **1122** being concave in a paraxial region thereof. The second lens element **1120** is made of plastic material and has the object-side surface **1121** and the image-side surface **1122** being both aspheric. The object-side surface **1121** of the second lens element **1120** has one inflection point.

(312) The third lens element **1130** with positive refractive power has an object-side surface **1131** being convex in a paraxial region thereof and an image-side surface **1132** being convex in a paraxial region thereof. The third lens element **1130** is made of plastic material and has the object-side surface **1131** and the image-side surface **1132** being both aspheric. The object-side surface **1131** of the third lens element **1130** has one inflection point. The image-side surface **1132** of the third lens element **1130** has two inflection points.

(313) The fourth lens element **1140** with negative refractive power has an object-side surface **1141** being concave in a paraxial region thereof and an image-side surface **1142** being concave in a paraxial region thereof. The fourth lens element **1140** is made of plastic material and has the object-side surface **1141** and the image-side surface **1142** being both aspheric. The object-side surface **1141** of the fourth lens element **1140** has one inflection point. The image-side surface **1142** of the fourth lens element **1140** has one inflection point.

(314) The fifth lens element **1150** with positive refractive power has an object-side surface **1151** being convex in a paraxial region thereof and an image-side surface **1152** being convex in a paraxial region thereof. The fifth lens element **1150** is made of plastic material and has the object-side surface **1151** and the image-side surface **1152** being both aspheric. The object-side surface **1151** of the fifth lens element **1150** has one inflection point. The image-side surface **1152** of the fifth lens element **1150** has one inflection point.

(315) The sixth lens element **1160** with negative refractive power has an object-side surface **1161** being concave in a paraxial region thereof and an image-side surface **1162** being concave in a paraxial region thereof. The sixth lens element **1160** is made of plastic material and has the object-side surface **1161** and the image-side surface **1162** being both aspheric. The object-side surface **1161** of the sixth lens element **1160** has one inflection point. The image-side surface **1162** of the sixth lens element **1160** has two inflection points.

(316) The seventh lens element **1170** with negative refractive power has an object-side surface **1171** being concave in a paraxial region thereof and an image-side surface **1172** being concave in a

paraxial region thereof. The seventh lens element **1170** is made of plastic material and has the object-side surface **1171** and the image-side surface **1172** being both aspheric. The object-side surface **1171** of the seventh lens element **1170** has one inflection point. The image-side surface **1172** of the seventh lens element **1170** has two inflection points. The image-side surface **1172** of the seventh lens element **1170** has at least one critical point in an off-axis region thereof.

(317) The eighth lens element **1180** with positive refractive power has an object-side surface **1181** being convex in a paraxial region thereof and an image-side surface **1182** being convex in a paraxial region thereof. The eighth lens element **1180** is made of plastic material and has the object-side surface **1181** and the image-side surface **1182** being both aspheric. The object-side surface **1181** of the eighth lens element **1180** has two inflection points. The image-side surface **1182** of the eighth lens element **1180** has one inflection point.

(318) The filter **1190** is made of glass material and located between the eighth lens element **1180** and the image surface **1195**, and will not affect the focal length of the imaging lens system. The image sensor **1199** is disposed on or near the image surface **1195** of the imaging lens system.

(319) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_2/N_2 , wherein an abbe number of the second lens element **1120** is V_2 , and a refractive index of the second lens element **1120** is N_2 .

(320) The detailed optical data of the 11th embodiment are shown in Table 21 and the aspheric surface data are shown in Table 22 below.

(321) TABLE-US-00030 TABLE 21 11th Embodiment $f = 6.05$ mm, $Fno = 1.56$, $HFOV = 19.0$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 2.041 (ASP) 1.300 Plastic 1.545 56.0 3.21 2 -9.390 (ASP) -0.092 3 Ape. Stop Plano 0.137 4 Lens 2 -9.933 (ASP) 0.170 Plastic 1.669 19.5 -7.46 5 10.087 (ASP) 0.235 6 Lens 3 7.810 (ASP) 0.244 Plastic 1.543 56.5 10.50 7 -20.922 (ASP) 0.255 8 Stop Plano 0.000 9 Lens 4 -23.137 (ASP) 0.253 Plastic 1.530 55.8 -5.97 10 3.682 (ASP) 0.611 11 Lens 5 5.154 (ASP) 0.180 Plastic 1.584 28.2 5.03 12 -6.749 (ASP) 0.069 13 Lens 6 -13.738 (ASP) 0.180 Plastic 1.634 23.8 -4.46 14 3.575 (ASP) 0.774 15 Lens 7 -355.132 (ASP) 0.170 Plastic 1.534 55.9 -4.73 16 2.546 (ASP) 0.067 17 Lens 8 11.545 (ASP) 0.553 Plastic 1.650 21.5 11.43 18 -20.482 (ASP) 0.300 19 Filter Plano 0.145 Glass 1.517 64.2 — 20 Plano 0.260 21 Image Plano — Note: Reference wavelength is 587.6 nm (d-line). An effective radius of the stop 1101 (Surface 8) is 1.265 mm.

(322) TABLE-US-00031 TABLE 22 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$ -6.3939E-01 2.1045E+01 -9.0000E+01 4.1077E+01 2.6345E+01 -2.2143E+01 $A_4 =$ 3.4326E-03 -8.2005E-03 -2.3078E-02 -2.3231E-02 9.9879E-03 1.0848E-01 $A_6 =$ 3.6745E-03 3.8905E-02 9.3193E-02 1.0350E-01 2.0998E-02 -6.4375E-02 $A_8 =$ -4.2343E-03 -3.3080E-02 -7.0499E-02 -9.2155E-02 -2.1418E-02 5.7529E-02 $A_{10} =$ 2.4145E-03 1.3681E-02 2.8016E-02 5.8750E-02 -8.6583E-03 -8.8155E-02 $A_{12} =$ -8.0969E-04 -2.8282E-03 -5.9948E-03 -2.3192E-02 2.2703E-02 7.6999E-02 $A_{14} =$ 9.6298E-05 2.4269E-04 5.7720E-04 3.7325E-03 -7.7143E-03 -2.1184E-02 Surface # 9 10 11 12 13 14 $k =$ -9.0000E+01 -1.5896E+01 -2.5589E+01 7.5808E+00 7.6794E+01 -8.2800E+01 $A_4 =$ 5.2790E-02 -5.2213E-02 -1.1178E-01 1.4353E-01 4.1053E-02 -1.4414E-02 $A_6 =$ 3.0264E-03 3.7691E-02 -1.8788E-01 -2.5331E-01 -4.4534E-02 -1.6864E-01 $A_8 =$ -3.3395E-02 -3.4861E-02 5.7656E-02 -8.3746E-02 -1.4308E-01 2.0387E-01 $A_{10} =$ 3.1214E-02 5.3918E-03 5.6074E-03 3.7276E-01 2.4471E-01 -1.3827E-01 $A_{12} =$ -7.3635E-03 1.3839E-03 2.8218E-02 -2.5003E-01 -1.6218E-01 4.9055E-02 $A_{14} =$ — — -1.4163E-02 5.8622E-02 4.0037E-02 -6.5344E-03 Surface # 15 16 17 18 $k =$ -9.0000E+01 1.1562E+00 2.5391E+01 -8.2360E+01 $A_4 =$ -5.0275E-01 -4.2419E-01 -1.8509E-02 -1.4277E-01 $A_6 =$ 2.6023E-01 3.0687E-01 1.2189E-02 1.6314E-01 $A_8 =$ 9.1309E-02 -2.3324E-01 -2.6423E-02 -1.0759E-01 $A_{10} =$ -3.4470E-01 1.4395E-01 1.8001E-02 3.8397E-02 $A_{12} =$ 3.4740E-01 -5.5715E-02 -5.5858E-03

-7.5337E-03 A14= -1.7004E-01 1.0895E-02 8.7325E-04 8.3110E-04 A16= 3.2302E-02
-8.1838E-04 -5.6634E-05 -4.4225E-05

(323) In the 11th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 11th embodiment, so an explanation in this regard will not be provided again.

(324) Moreover, these parameters can be calculated from Table 21 and Table 22 as the following values and satisfy the following conditions:

(325) TABLE-US-00032 11th Embodiment f [mm] 6.05 EPD/BL 5.50 Fno 1.56 f/EPD 1.56 HFOV [deg.] 19.0 (Y11 - ImgH)/EPD -0.05 V8 21.5 $f \times \tan(\text{HFOV})/\text{EPD}$ 0.54 Vmin 19.5 ($|Y11 - Y82| + \text{BL}$)/f 0.12 ΣVi 317.2 (Y82 - Y51)/(Y11 - Y51) 0.96 Nmax 1.669 ($|Y11 - Y82| + |Y82 - \text{ImgH}|$)/f 0.04 (Vi/Ni)min 11.66 f/ImgH 2.83 $\Sigma \text{CT}/\Sigma \text{AT}$ 1.48 Yc72/f 0.10 R1/CT1 1.57 $|f/R1| + |f/R2|$ 3.61 BL/CT1 0.54 $|f/R3| + |f/R4|$ 1.21 R8/f 0.61 $|f/R5| + |f/R6|$ 1.06 (R11 + R12)/ 0.59 $|f/R7| + |f/R8|$ 1.91 (R11 - R12) (R13 - R14)/ 1.01 $|f/R9| + |f/R10|$ 2.07 (R13 + R14) f/f1 1.89 $|f/R11| + |f/R12|$ 2.13 f/f5 1.20 $|f/R13| + |f/R14|$ 2.40 f5/f7 -1.06 $|f/R15| + |f/R16|$ 0.82 f8/f5 2.27 V1/N1 36.27 f/f5 + f/f6 -0.15 V2/N2 11.66 ImgH [mm] 2.14 V3/N3 36.60 TL [mm] 5.81 V4/N4 36.45 TL/f 0.96 V5/N5 17.80 TL/EPD 1.50 V6/N6 14.59 SL/TL 0.79 V7/N7 36.46 BL/TD 0.14 V8/N8 13.01 EPD/ImgH 1.81 — —

12th Embodiment

(326) FIG. 23 is a schematic view of an image capturing unit according to the 12th embodiment of the present disclosure. FIG. 24 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 12th embodiment. In FIG. 23, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 1299. The imaging lens system includes, in order from an object side to an image side, a first lens element 1210, an aperture stop 1200, a second lens element 1220, a third lens element 1230, a stop 1201, a fourth lens element 1240, a fifth lens element 1250, a sixth lens element 1260, a seventh lens element 1270, an eighth lens element 1280, a filter 1290 and an image surface 1295. The imaging lens system includes eight lens elements (1210, 1220, 1230, 1240, 1250, 1260, 1270 and 1280) with no additional lens element disposed between each of the adjacent eight lens elements.

(327) The first lens element 1210 with positive refractive power has an object-side surface 1211 being convex in a paraxial region thereof and an image-side surface 1212 being convex in a paraxial region thereof. The first lens element 1210 is made of plastic material and has the object-side surface 1211 and the image-side surface 1212 being both aspheric. The object-side surface 1211 of the first lens element 1210 has two inflection points. The image-side surface 1212 of the first lens element 1210 has one inflection point.

(328) The second lens element 1220 with negative refractive power has an object-side surface 1221 being concave in a paraxial region thereof and an image-side surface 1222 being concave in a paraxial region thereof. The second lens element 1220 is made of plastic material and has the object-side surface 1221 and the image-side surface 1222 being both aspheric. The object-side surface 1221 of the second lens element 1220 has one inflection point.

(329) The third lens element 1230 with positive refractive power has an object-side surface 1231 being convex in a paraxial region thereof and an image-side surface 1232 being convex in a paraxial region thereof. The third lens element 1230 is made of plastic material and has the object-side surface 1231 and the image-side surface 1232 being both aspheric. The object-side surface 1231 of the third lens element 1230 has one inflection point. The image-side surface 1232 of the third lens element 1230 has one inflection point.

(330) The fourth lens element 1240 with negative refractive power has an object-side surface 1241 being concave in a paraxial region thereof and an image-side surface 1242 being concave in a

paraxial region thereof. The fourth lens element **1240** is made of plastic material and has the object-side surface **1241** and the image-side surface **1242** being both aspheric. The object-side surface **1241** of the fourth lens element **1240** has one inflection point. The image-side surface **1242** of the fourth lens element **1240** has one inflection point.

(331) The fifth lens element **1250** with positive refractive power has an object-side surface **1251** being convex in a paraxial region thereof and an image-side surface **1252** being concave in a paraxial region thereof. The fifth lens element **1250** is made of plastic material and has the object-side surface **1251** and the image-side surface **1252** being both aspheric. The object-side surface **1251** of the fifth lens element **1250** has one inflection point. The image-side surface **1252** of the fifth lens element **1250** has two inflection points.

(332) The sixth lens element **1260** with negative refractive power has an object-side surface **1261** being convex in a paraxial region thereof and an image-side surface **1262** being concave in a paraxial region thereof. The sixth lens element **1260** is made of plastic material and has the object-side surface **1261** and the image-side surface **1262** being both aspheric. The object-side surface **1261** of the sixth lens element **1260** has two inflection points. The image-side surface **1262** of the sixth lens element **1260** has one inflection point.

(333) The seventh lens element **1270** with negative refractive power has an object-side surface **1271** being convex in a paraxial region thereof and an image-side surface **1272** being concave in a paraxial region thereof. The seventh lens element **1270** is made of plastic material and has the object-side surface **1271** and the image-side surface **1272** being both aspheric. The object-side surface **1271** of the seventh lens element **1270** has two inflection points. The image-side surface **1272** of the seventh lens element **1270** has two inflection points. The image-side surface **1272** of the seventh lens element **1270** has at least one critical point in an off-axis region thereof.

(334) The eighth lens element **1280** with positive refractive power has an object-side surface **1281** being convex in a paraxial region thereof and an image-side surface **1282** being convex in a paraxial region thereof. The eighth lens element **1280** is made of plastic material and has the object-side surface **1281** and the image-side surface **1282** being both aspheric. The object-side surface **1281** of the eighth lens element **1280** has two inflection points. The image-side surface **1282** of the eighth lens element **1280** has two inflection points.

(335) The filter **1290** is made of glass material and located between the eighth lens element **1280** and the image surface **1295**, and will not affect the focal length of the imaging lens system. The image sensor **1299** is disposed on or near the image surface **1295** of the imaging lens system.

(336) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to V_2/N_2 , wherein an abbe number of the second lens element **1220** is V_2 , and a refractive index of the second lens element **1220** is N_2 .

(337) The detailed optical data of the 12th embodiment are shown in Table 23 and the aspheric surface data are shown in Table 24 below.

(338) TABLE-US-00033 TABLE 23 12th Embodiment $f = 6.30$ mm. $F_{no} = 1.57$, HFOV = 18.0 deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 2.045 (ASP) 1.300 Plastic 1.545 56.0 3.21 2 -9.352 (ASP) -0.088 3 Ape. Stop Plano 0.133 4 Lens 2 -10.115 (ASP) 0.170 Plastic 1.669 19.5 -7.21 5 9.282 (ASP) 0.101 6 Lens 3 7.434 (ASP) 0.368 Plastic 1.543 56.5 7.54 7 -8.957 (ASP) 0.219 8 Stop Plano 0.100 9 Lens 4 -19.874 (ASP) 0.203 Plastic 1.530 55.8 -4.79 10 2.925 (ASP) 0.604 11 Lens 5 5.865 (ASP) 0.180 Plastic 1.584 28.2 10.34 12 197.222 (ASP) 0.070 13 Lens 6 11.135 (ASP) 0.180 Plastic 1.639 23.5 -6.37 14 2.962 (ASP) 0.755 15 Lens 7 17.679 (ASP) 0.170 Plastic 1.534 55.9 -5.14 16 2.367 (ASP) 0.126 17 Lens 8 11.718 (ASP) 0.562 Plastic 1.650 21.5 9.80 18 -13.705 (ASP) 0.300 19 Filter Plano 0.145 Glass 1.517 64.2 — 20 Plano 0.258 21 Image Plano — Note: Reference wavelength is 587.6 nm (d-line). An effective radius of the stop 1201 (Surface 8) is 1.280 mm.

(339) TABLE-US-00034 TABLE 24 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$

-7.2960E-01 2.0042E+01 -8.9998E+01 3.3917E+01 2.3030E+01 -1.2027E+01 A4=
 4.2131E-03 -8.0505E-03 -6.4992E-03 -9.5777E-04 9.8111E-03 9.0978E-02 A6=
 9.8921E-04 3.8124E-02 4.4375E-02 -1.8482E-02 -8.7019E-02 -8.4469E-02 A8=
 -1.1799E-03 -3.1473E-02 -1.0248E-02 9.5767E-02 2.2144E-01 1.7437E-01 A10=
 4.6977E-04 1.2432E-02 -5.4784E-03 -6.7034E-02 -2.0648E-01 -2.0510E-01 A12=
 -2.9933E-04 -2.4747E-03 2.6123E-03 1.5874E-02 9.0424E-02 1.1584E-01 A14=
 4.9687E-05 2.0808E-04 -2.6269E-04 -9.4331E-04 -1.5235E-02 -2.3646E-02 Surface # 9 10
 11 12 13 14 k= -7.8475E+01 -1.4772E+01 -4.1217E+01 -8.3815E+01 -8.9975E+01
 -8.2800E+01 A4= 2.7303E-02 -4.7118E-02 -5.9512E-02 2.0184E-01 -8.9618E-02
 -2.8612E-02 A6= 6.7571E-02 5.1557E-02 -3.0712E-01 -2.3946E-01 4.9576E-01
 -1.1070E-01 A8= -9.4708E-02 -7.2384E-02 -2.2661E-01 -5.9978E-01 -1.0636E+00
 3.0515E-01 A10= 6.0013E-02 3.8975E-02 8.5899E-01 1.2892E+00 1.0479E+00
 -3.6683E-01 A12= -1.2726E-02 -8.4857E-03 -6.8394E-01 -8.9050E-01 -5.4003E-01
 1.9818E-01 A14= — — 1.8330E-01 2.2466E-01 1.1892E-01 -3.9744E-02 Surface # 15 16
 17 18 k= 4.4914E+01 1.0265E+00 2.5391E+01 -8.2360E+01 A4= -5.1902E-01
 -4.9670E-01 -8.1576E-02 -1.7771E-01 A6= 1.7449E-01 3.4730E-01 1.0867E-01
 1.7737E-01 A8= 2.4379E-01 -2.3304E-01 -1.1261E-01 -1.0509E-01 A10= -4.8573E-01
 1.2486E-01 6.2218E-02 3.0738E-02 A12= 4.5099E-01 -4.3112E-02 -1.8244E-02
 -3.5083E-03 A14= -2.2645E-01 7.2301E-03 2.7700E-03 -2.9503E-05 A16= 4.5830E-02
 -4.2224E-04 -1.7382E-04 2.1497E-05

(340) In the 12th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 12th embodiment, so an explanation in this regard will not be provided again.

(341) Moreover, these parameters can be calculated from Table 23 and Table 24 as the following values and satisfy the following conditions:

(342) TABLE-US-00035 12th Embodiment f [mm] 6.30 EPD/BL 5.70 Fno 1.57 f/EPD 1.57 HFOV [deg.] 18.0 (Y11 - ImgH)/EPD -0.02 V8 21.5 f × tan(HFOV)/EPD 0.51 Vmin 19.5 (|Y11 - Y82| + BL)/f 0.13 ΣVi 316.8 (Y82 - Y51)/(Y11 - Y51) 0.89 Nmax 1.669 (|Y11 - Y82| + |Y82 - ImgH|)/f 0.05 (Vi/Ni)min 11.66 f/ImgH 3.00 ΣCT/ΣAT 1.55 Yc72/f 0.09 R1/CT1 1.57 |f/R1| + |f/R2| 3.75 BL/CT1 0.54 |f/R3| + |f/R4| 1.30 R8/f 0.46 |f/R5| + |f/R6| 1.55 (R11 + R12)/ 1.72 |f/R7| + |f/R8| 2.47 (R11 - R12) (R13 - R14)/ 0.76 |f/R9| + |f/R10| 1.11 (R13 + R14) f/f1 1.96 |f/R11| + |f/R12| 2.69 f/f5 0.61 |f/R13| + |f/R14| 3.02 f5/f7 -2.01 |f/R15| + |f/R16| 1.00 f8/f5 0.95 V1/N1 36.27 f/f5 + f/f6 -0.38 V2/N2 11.66 ImgH [mm] 2.10 V3/N3 36.60 TL [mm] 5.86 V4/N4 36.45 TL/f 0.93 V5/N5 17.80 TL/EPD 1.46 V6/N6 14.34 SL/TL 0.79 V7/N7 36.46 BL/TD 0.14 V8/N8 13.01 EPD/ImgH 1.91 — —

13th Embodiment

(343) FIG. 25 is a schematic view of an image capturing unit according to the 13th embodiment of the present disclosure. FIG. 26 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 13th embodiment. In FIG. 25, the image capturing unit includes the imaging lens system (its reference numeral is omitted) of the present disclosure and an image sensor 1399. The imaging lens system includes, in order from an object side to an image side, a first lens element 1310, an aperture stop 1300, a second lens element 1320, a third lens element 1330, a stop 1301, a fourth lens element 1340, a fifth lens element 1350, a sixth lens element 1360, a seventh lens element 1370, an eighth lens element 1380, a filter 1390 and an image surface 1395. The imaging lens system includes eight lens elements (1310, 1320, 1330, 1340, 1350, 1360, 1370 and 1380) with no additional lens element disposed between each of the adjacent eight lens elements.

(344) The first lens element 1310 with positive refractive power has an object-side surface 1311

being convex in a paraxial region thereof and an image-side surface **1312** being convex in a paraxial region thereof. The first lens element **1310** is made of plastic material and has the object-side surface **1311** and the image-side surface **1312** being both aspheric. The image-side surface **1312** of the first lens element **1310** has one inflection point.

(345) The second lens element **1320** with negative refractive power has an object-side surface **1321** being convex in a paraxial region thereof and an image-side surface **1322** being concave in a paraxial region thereof. The second lens element **1320** is made of plastic material and has the object-side surface **1321** and the image-side surface **1322** being both aspheric. The object-side surface **1321** of the second lens element **1320** has one inflection point.

(346) The third lens element **1330** with negative refractive power has an object-side surface **1331** being convex in a paraxial region thereof and an image-side surface **1332** being concave in a paraxial region thereof. The third lens element **1330** is made of plastic material and has the object-side surface **1331** and the image-side surface **1332** being both aspheric. The image-side surface **1332** of the third lens element **1330** has one inflection point.

(347) The fourth lens element **1340** with positive refractive power has an object-side surface **1341** being convex in a paraxial region thereof and an image-side surface **1342** being concave in a paraxial region thereof. The fourth lens element **1340** is made of plastic material and has the object-side surface **1341** and the image-side surface **1342** being both aspheric. The image-side surface **1342** of the fourth lens element **1340** has one inflection point.

(348) The fifth lens element **1350** with positive refractive power has an object-side surface **1351** being convex in a paraxial region thereof and an image-side surface **1352** being convex in a paraxial region thereof. The fifth lens element **1350** is made of plastic material and has the object-side surface **1351** and the image-side surface **1352** being both aspheric. The object-side surface **1351** of the fifth lens element **1350** has one inflection point. The image-side surface **1352** of the fifth lens element **1350** has one inflection point.

(349) The sixth lens element **1360** with negative refractive power has an object-side surface **1361** being concave in a paraxial region thereof and an image-side surface **1362** being concave in a paraxial region thereof. The sixth lens element **1360** is made of plastic material and has the object-side surface **1361** and the image-side surface **1362** being both aspheric. The object-side surface **1361** of the sixth lens element **1360** has two inflection points. The image-side surface **1362** of the sixth lens element **1360** has one inflection point.

(350) The seventh lens element **1370** with negative refractive power has an object-side surface **1371** being concave in a paraxial region thereof and an image-side surface **1372** being convex in a paraxial region thereof. The seventh lens element **1370** is made of plastic material and has the object-side surface **1371** and the image-side surface **1372** being both aspheric. The object-side surface **1371** of the seventh lens element **1370** has one inflection point. The image-side surface **1372** of the seventh lens element **1370** has three inflection points.

(351) The eighth lens element **1380** with negative refractive power has an object-side surface **1381** being convex in a paraxial region thereof and an image-side surface **1382** being concave in a paraxial region thereof. The eighth lens element **1380** is made of plastic material and has the object-side surface **1381** and the image-side surface **1382** being both aspheric. The object-side surface **1381** of the eighth lens element **1380** has two inflection points. The image-side surface **1382** of the eighth lens element **1380** has two inflection points.

(352) The filter **1390** is made of glass material and located between the eighth lens element **1380** and the image surface **1395**, and will not affect the focal length of the imaging lens system. The image sensor **1399** is disposed on or near the image surface **1395** of the imaging lens system.

(353) In this embodiment, an Abbe number of the i-th lens element is V_i , a refractive index of the i-th lens element is N_i , a minimum value of V_i/N_i is $(V_i/N_i)_{\min}$, and $(V_i/N_i)_{\min}$ is equal to $V8/N8$, wherein an abbe number of the eighth lens element **1380** is $V8$, and a refractive index of the eighth lens element **1380** is $N8$.

(354) The detailed optical data of the 13th embodiment are shown in Table 25 and the aspheric surface data are shown in Table 26 below.

(355) TABLE-US-00036 TABLE 25 13th Embodiment $f = 5.63$ mm, $Fno = 1.48$, $HFOV = 20.2$ deg. Surface # Curvature Radius Thickness Material Index Abbe # Focal Length 0 Object Plano Infinity 1 Lens 1 2.066 (ASP) 1.264 Plastic 1.545 56.0 3.37 2 -12.844 (ASP) -0.013 3 Ape. Stop Plano 0.053 4 Lens 2 11.975 (ASP) 0.170 Plastic 1.650 21.5 -8.98 5 3.904 (ASP) 0.309 6 Lens 3 8.996 (ASP) 0.160 Plastic 1.582 30.2 -7.25 7 2.856 (ASP) 0.450 8 Stop Plano -0.400 9 Lens 4 2.402 (ASP) 0.565 Plastic 1.530 55.8 13.52 10 3.317 (ASP) 0.739 11 Lens 5 14.137 (ASP) 0.203 Plastic 1.584 28.2 7.13 12 -5.873 (ASP) 0.035 13 Lens 6 -14.090 (ASP) 0.409 Plastic 1.614 26.0 -9.62 14 10.269 (ASP) 0.648 15 Lens 7 -3.258 (ASP) 0.180 Plastic 1.534 55.9 -7.41 16 -18.793 (ASP) 0.040 17 Lens 8 12.127 (ASP) 0.389 Plastic 1.669 19.5 -84.72 18 9.861 (ASP) 0.300 19 Filter Plano 0.145 Glass 1.517 64.2 — 20 Plano 0.257 21 Image Plano — Note: Reference wavelength is 587.6 nm (d-line). An effective radius of the stop 1301 (Surface 8) is 1.311 mm.

(356) TABLE-US-00037 TABLE 26 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k =$
-5.3196E-01 4.1035E+01 4.8091E+01 -1.5333E+01 -1.6409E+01 -6.2126E+00 $A4 =$
3.1358E-03 1.7459E-02 -7.1320E-02 -7.6403E-02 4.8029E-04 1.4696E-01 $A6 =$
3.5379E-03 2.9914E-03 1.3716E-01 1.8245E-01 7.8521E-02 -7.5349E-02 $A8 =$
-3.6714E-03 -7.8525E-03 -9.5564E-02 -1.0486E-01 -1.2843E-01 -4.6673E-02 $A10 =$
1.9702E-03 4.2343E-03 3.3222E-02 1.9384E-02 8.5984E-02 7.8585E-02 $A12 =$
-5.8353E-04 -9.8604E-04 -4.5860E-03 5.0736E-03 -2.6528E-02 -3.1847E-02 $A14 =$
6.4154E-05 9.1477E-05 5.3819E-06 -1.4840E-03 3.4764E-03 3.8198E-03 Surface # 9 10 11
12 13 14 $k =$ 5.5612E-01 -2.2694E+00 -3.8660E+00 1.1748E+01 5.9898E+01 -8.2800E+01
 $A4 =$ 4.8286E-02 -5.1856E-02 -7.8773E-02 4.8491E-02 -3.4647E-02 -1.2983E-01 $A6 =$
-8.1591E-02 -5.5715E-03 -3.8407E-02 -6.8719E-02 1.5665E-02 6.9287E-02 $A8 =$
7.0165E-02 2.6842E-02 -2.9826E-01 -1.9097E-01 3.0222E-02 -4.2653E-02 $A10 =$
-2.1655E-02 -2.1859E-02 3.3801E-01 2.7884E-01 -6.0168E-02 1.5587E-03 $A12 =$
1.8103E-03 5.0126E-03 -1.2889E-01 -1.0422E-01 3.9326E-02 6.9947E-03 $A14 =$ — —
1.6785E-02 1.0352E-02 -8.5761E-03 -1.6368E-03 Surface # 15 16 17 18 $k =$ -2.8287E+01
1.7904E+01 3.5002E+01 -8.2360E+01 $A4 =$ -1.7535E-01 1.5339E-01 6.7850E-02
-1.0542E-01 $A6 =$ -6.6445E-02 -4.4875E-01 -1.5383E-01 1.1855E-01 $A8 =$ 2.8817E-01
4.7524E-01 7.6537E-02 -1.1744E-01 $A10 =$ -2.9961E-01 -2.8583E-01 -1.1987E-02
6.1765E-02 $A12 =$ 1.4584E-01 9.9748E-02 -1.1173E-03 -1.6539E-02 $A14 =$ -3.3394E-02
-1.8732E-02 5.3377E-04 2.2172E-03 $A16 =$ 2.9151E-03 1.4646E-03 -4.5657E-05
-1.2023E-04

(357) In the 13th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 13th embodiment, so an explanation in this regard will not be provided again.

(358) Moreover, these parameters can be calculated from Table 25 and Table 26 as the following values and satisfy the following conditions:

(359) TABLE-US-00038 13th Embodiment f [mm] 5.63 EPD/BL 5.41 Fno 1.48 f/EPD 1.48 $HFOV$ [deg.] 20.2 $(Y11 - ImgH)/EPD$ -0.05 $V8$ 19.5 $f \times \tan(HFOV)/EPD$ 0.54 $Vmin$ 19.5 $(|Y11 - Y82| + BL)/f$ 0.13 ΣVi 293.1 $(Y82 - Y51)/(Y11 - Y51)$ 0.97 $Nmax$ 1.669 $(|Y11 - Y82| + |Y82 - ImgH|)/f$ 0.04 $(Vi/Ni)min$ 11.66 $f/ImgH$ 2.63 $\Sigma CT/\Sigma AT$ 1.79 $Yc72/f$ — $R1/CT1$ 1.63 $|f/R1| + |f/R2|$ 3.16 $BL/CT1$ 0.56 $|f/R3| + |f/R4|$ 1.91 $R8/f$ 0.59 $|f/R5| + |f/R6|$ 2.60 $(R11 + R12)/0.16$ $|f/R7| + |f/R8|$ 4.04 $(R11 - R12)(R13 - R14)/-0.70$ $|f/R9| + |f/R10|$ 1.36 $(R13 + R14)/f$ 1.67 $|f/R11| + |f/R12|$ 0.95 $f/f5$ 0.79 $|f/R13| + |f/R14|$ 2.03 $f5/f7$ -0.96 $|f/R15| + |f/R16|$ 1.03 $f8/f5$ -11.88 $V1/N1$ 36.27 $f/f5 + f/f6$ 0.20 $V2/N2$ 13.01 $ImgH$ [mm] 2.14 $V3/N3$ 19.11 TL [mm] 5.90 $V4/N4$ 36.45 TL/f 1.05 $V5/N5$ 17.80 TL/EPD 1.55 $V6/N6$ 16.09 SL/TL 0.79 $V7/N7$ 36.46 BL/TD 0.14 $V8/N8$ 11.66 $EPD/ImgH$

14th Embodiment

(360) FIG. 27 is a perspective view of an image capturing unit according to the 14th embodiment of the present disclosure. In this embodiment, an image capturing unit **10** is a camera module including a lens unit **11**, a driving device **12**, an image sensor **13** and an image stabilizer **14**. The lens unit **11** includes the imaging lens system of the present disclosure, a barrel and a holder member (their reference numerals are omitted) for holding the imaging lens system. The imaging light converges in the lens unit **11** of the image capturing unit **10** to generate an image with the driving device **12** utilized for image focusing on the image sensor **13**, and the generated image is then digitally transmitted to other electronic component for further processing.

(361) The driving device **12** can have auto focusing functionality, and different driving configurations can be obtained through the usages of voice coil motors (VCM), micro electro-mechanical systems (MEMS), piezoelectric systems, or shape memory alloy materials. The driving device **12** is favorable for obtaining a better imaging position of the lens unit **11**, so that a clear image of the imaged object can be captured by the lens unit **11** with different object distances. The image sensor **13** (for example, CCD or CMOS), which can feature high photosensitivity and low noise, is disposed on the image surface of the imaging lens system to provide higher image quality.

(362) The image stabilizer **14**, such as an accelerometer, a gyro sensor and a Hall Effect sensor, is configured to work with the driving device **12** to provide optical image stabilization (OIS). The driving device **12** working with the image stabilizer **14** is favorable for compensating for pan and tilt of the lens unit **11** to reduce blurring associated with motion during exposure. In some cases, the compensation can be provided by electronic image stabilization (EIS) with image processing software, thereby improving image quality while in motion or low-light conditions.

15th Embodiment

(363) FIG. 28 is one perspective view of an electronic device according to the 15th embodiment of the present disclosure. FIG. 29 is another perspective view of the electronic device in FIG. 28. FIG. 30 is a block diagram of the electronic device in FIG. 28.

(364) In this embodiment, an electronic device **20** is a smartphone including the image capturing unit **10** disclosed in the 14th embodiment, an image capturing unit **10a**, an image capturing unit **10b**, an image capturing unit **10c**, a flash module **21**, a focus assist module **22**, an image signal processor **23**, a user interface **24** and an image software processor **25**. The image capturing unit **10c** is located on the same side as the user interface **24**, and the image capturing unit **10**, the image capturing unit **10a** and the image capturing unit **10b** are located on the opposite side. The image capturing unit **10**, the image capturing unit **10a** and the image capturing unit **10b** all face the same side. Furthermore, the image capturing unit **10a**, the image capturing unit **10b** and the image capturing unit **10c** all have a configuration similar to that of the image capturing unit **10**. In detail, each of the image capturing unit **10a**, the image capturing unit **10b** and the image capturing unit **10c** includes a lens unit, a driving device, an image sensor and an image stabilizer, and the lens unit includes an optical lens assembly, a barrel and a holder member for holding the optical lens assembly.

(365) In this embodiment, the image capturing unit **10** is a telephoto image capturing unit, the image capturing unit **10a** is a wide-angle image capturing unit and the image capturing unit **10b** is a standard image capturing unit having a field of view ranging between that of the image capturing unit **10** and the image capturing unit **10a**. In addition, the field of view of the image capturing unit **10** and the field of view of the image capturing unit **10a** are different from each other by at least 20 degrees. The image capturing units **10**, **10a**, **10b** have different fields of view, such that the electronic device **20** has various magnification ratios so as to meet the requirement of optical zoom functionality. In this embodiment, the electronic device **20** includes multiple image capturing units **10**, **10a**, **10b** and **10c**, but the present disclosure is not limited to the number of image capturing units.

(366) When a user captures images of an object **26**, the light rays converge in the image capturing unit **10**, the image capturing unit **10a** or the image capturing unit **10b** to generate an image(s), and the flash module **21** is activated for light supplement. The focus assist module **22** detects the object distance of the imaged object **26** to achieve fast auto focusing. The image signal processor **23** is configured to optimize the captured image to improve image quality. The light beam emitted from the focus assist module **22** can be either conventional infrared or laser. In addition, the electronic device **20** can capture images of the object **26** via the image capturing unit **10c**. The user interface **24** can be a touch screen or a physical button. The user is able to interact with the user interface **24** and the image software processor **25** having multiple functions to capture images and complete image processing. The image processed by the image software processor **25** can be displayed on the user interface **24**.

(367) The smartphone in this embodiment is only exemplary for showing the image capturing unit **10** of the present disclosure installed in an electronic device, and the present disclosure is not limited thereto. The image capturing unit **10** can be optionally applied to optical systems with a movable focus. Furthermore, the imaging lens system of the image capturing unit **10** features good capability in aberration corrections and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart televisions, network surveillance devices, dashboard cameras, vehicle backup cameras, multi-camera devices, image recognition systems, motion sensing input devices, wearable devices and other electronic imaging devices.

(368) The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-26 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

Claims

1. An imaging lens system comprising a total of eight lens elements, the eight lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, and each of the eight lens elements having an object-side surface facing toward the object side and an image-side surface facing toward the image side; wherein the fifth lens element has positive refractive power, the image-side surface of the fourth lens element is concave in a paraxial region thereof, the object-side surface of the fifth lens element is convex in a paraxial region thereof, the object-side surface of the sixth lens element is concave in a paraxial region thereof, the image-side surface of the eighth lens element is concave in a paraxial region thereof, and at least one of the eight lens elements has at least one lens surface having at least one inflection point; wherein a central thickness of the eighth lens element is larger than a central thickness of the second lens element; wherein a minimum value among Abbe numbers of all lens elements of the imaging lens system is $V_{\min}$, half of a maximum field of view of the imaging lens system is HFOV, a focal length of the imaging lens system is f , an entrance pupil diameter of the imaging lens system is EPD, and the following conditions are satisfied:

$10 < V_{\min} < 20.0$;

$5.0 \text{ degrees} < \text{HFOV} < 25.0 \text{ degrees}$; and

$0.20 < f \times \tan(\text{HFOV}) / \text{EPD} < 0.73$.

2. The imaging lens system of claim 1, wherein the first lens element has positive refractive power, the object-side surface of the first lens element is convex in a paraxial region thereof, and each of at least three lens elements of the imaging lens system has at least one lens surface having at least one inflection point.
3. The imaging lens system of claim 1, wherein at least four lens elements of the imaging lens system are made of plastic material; wherein a curvature radius of the image-side surface of the fourth lens element is $R8$, the focal length of the imaging lens system is f , and the following condition is satisfied:
 $-1.50 < R8/f < 1.50$.
4. The imaging lens system of claim 1, wherein a curvature radius of the object-side surface of the seventh lens element is $R13$, a curvature radius of the image-side surface of the seventh lens element is $R14$, and the following condition is satisfied:
 $-2.0 < (R13 - R14)/(R13 + R14) < 3.0$.
5. The imaging lens system of claim 1, wherein a central thickness of the sixth lens element is larger than a central thickness of the seventh lens element.
6. The imaging lens system of claim 1, wherein each of at least four lens elements of the imaging lens system has an Abbe number smaller than 30.0.
7. The imaging lens system of claim 1, wherein a maximum effective radius of the object-side surface of the first lens element is $Y11$, a maximum effective radius of the image-side surface of the eighth lens element is $Y82$, a maximum image height of the imaging lens system is $ImgH$, the focal length of the imaging lens system is f , and the following condition is satisfied:
 $0 < (|Y11 - Y82| + |Y82 - ImgH|)/f < 0.15$.
8. The imaging lens system of claim 1, wherein a maximum effective radius of the object-side surface of the first lens element is $Y11$, a maximum effective radius of the image-side surface of the eighth lens element is $Y82$, an axial distance between the image-side surface of the eighth lens element and an image surface is BL , the focal length of the imaging lens system is f , and the following condition is satisfied: $0 < (|Y11 - Y82| + BL)/f < 0.20$.
9. The imaging lens system of claim 1, wherein an absolute value of a curvature radius of the image-side surface of the first lens element is larger than an absolute value of a curvature radius of the object-side surface of the eighth lens element.
10. The imaging lens system of claim 1, wherein an axial distance between the fourth lens element and the fifth lens element is larger than an axial distance between the seventh lens element and the eighth lens element.
11. The imaging lens system of claim 1, wherein an axial distance between the sixth lens element and the seventh lens element is larger than an axial distance between the third lens element and the fourth lens element.
12. An image capturing unit comprising: the imaging lens system of claim 1; and an image sensor disposed on an image surface of the imaging lens system.
13. An electronic device comprising at least two image capturing units, wherein the at least two image capturing units face a same side and comprise: a first image capturing unit comprising the imaging lens system of claim 1 and a first image sensor, wherein the first image sensor is disposed on an image surface of the imaging lens system; and a second image capturing unit comprising an optical lens assembly and a second image sensor, wherein the second image sensor is disposed on an image surface of the optical lens assembly.
14. An imaging lens system comprising a total of eight lens elements, the eight lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, and each of the eight lens elements having an object-side surface facing toward the object side and an image-side surface facing toward the image side; wherein the image-side surface of the second lens element is convex in a paraxial region thereof,

the object-side surface of the third lens element is concave in a paraxial region thereof, the image-side surface of the fourth lens element is concave in a paraxial region thereof, the object-side surface of the sixth lens element is concave in a paraxial region thereof, the object-side surface of the eighth lens element is convex in a paraxial region thereof, and at least one lens element of the imaging lens system has at least one lens surface having at least one inflection point; wherein a central thickness of the sixth lens element is larger than a central thickness of the second lens element; wherein a maximum effective radius of the object-side surface of the first lens element is Y11, a maximum effective radius of the image-side surface of the eighth lens element is Y82, an axial distance between the image-side surface of the eighth lens element and an image surface is BL, a focal length of the imaging lens system is f, and the following condition is satisfied:
 $0 < (|Y11 - Y82| + BL) / f < 0.20$.

15. The imaging lens system of claim 14, wherein half of a maximum field of view of the imaging lens system is HFOV, and the following condition is satisfied:

$5.0 \text{ degrees} < \text{HFOV} < 25.0 \text{ degrees}$.

16. The imaging lens system of claim 14, wherein a minimum value among Abbe numbers of all lens elements of the imaging lens system is Vmin, and the following condition is satisfied:

$8.0 < V_{\min} < 22.5$.

17. The imaging lens system of claim 14, wherein the object-side surface of the first lens element is convex in a paraxial region thereof, and each of at least two lens elements of the imaging lens system has at least one lens surface having at least one inflection point.

18. The imaging lens system of claim 14, wherein at least four lens elements of the imaging lens system are made of plastic material; and wherein the axial distance between the image-side surface of the eighth lens element and the image surface is BL, an axial distance between the object-side surface of the first lens element and the image-side surface of the eighth lens element is TD, and the following condition is satisfied:

$0 < BL / TD < 0.18$.

19. The imaging lens system of claim 14, wherein the focal length of the imaging lens system is f, half of a maximum field of view of the imaging lens system is HFOV, an entrance pupil diameter of the imaging lens system is EPD, and the following condition is satisfied:

$0.20 < f \times \tan(\text{HFOV}) / \text{EPD} < 1.0$.

20. The imaging lens system of claim 14, wherein the focal length of the imaging lens system is f, a focal length of the fifth lens element is f5, and the following condition is satisfied:

$-0.25 < f / f5 < 1.50$.

21. The imaging lens system of claim 14, wherein a central thickness of the eighth lens element is larger than an axial distance between the seventh lens element and the eighth lens element; wherein a curvature radius of the object-side surface of the seventh lens element is R13, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$-2.0 < (R13 - R14) / (R13 + R14) < 3.0$.

22. The imaging lens system of claim 14, wherein each of at least four lens elements of the imaging lens system has an Abbe number smaller than 35.0.

23. The imaging lens system of claim 14, wherein an Abbe number of the first lens element is V1, an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, an Abbe number of the fifth lens element is V5, an Abbe number of the sixth lens element is V6, an Abbe number of the seventh lens element is V7, an Abbe number of the eighth lens element is V8, an Abbe number of the i-th lens element is Vi, a maximum image height of the imaging lens system is ImgH, and the following conditions are satisfied:

$150.0 < \sum Vi < 320.0$, wherein $i = 1, 2, 3, 4, 5, 6, 7, 8$; and

$1.50 \text{ mm} < \text{ImgH} < 4.50 \text{ mm}$.

24. The imaging lens system of claim 14, wherein the maximum effective radius of the object-side surface of the first lens element is Y_{11} , the maximum effective radius of the image-side surface of the eighth lens element is Y_{82} , a maximum image height of the imaging lens system is $ImgH$, the focal length of the imaging lens system is f , and the following condition is satisfied:

$$0 < (|Y_{11} - Y_{82}| + |Y_{82} - ImgH|) / f < 0.15.$$
